

APPENDIX

This appendix provides supplementary technical details, mathematical derivations, implementation descriptions, and extended experimental results to support the main findings of the paper. The content is organized as follows:

- **App. A: Task Specification.** Defines the common kinematic variables, orientation quantities, control symbols, reward terms, task settings, observations, initialization protocols, and termination criteria for the full musculoskeletal humanoid benchmark.
- **App. B: Evaluation Details.** Describes the seven metric families: success rate, cumulative reward, peak-efficiency steps, robustness (action, observation, and dynamics perturbations), activation energy, joint smoothness, and EMG similarity.
- **Apps. C–D: Comparison Scope and Aggregation.** Specifies the evaluated task subsets and defines the descriptive summaries of success, coverage, and perturbation robustness.
- **App. E: Detailed Descriptions of Reward-Based RL Baselines.** Provides detailed explanations of the pure reinforcement learning baselines, including PPO, SAC, Dep-RL, and DynSyn-SAC, with emphasis on their exploration mechanisms and suitability for high-dimensional musculoskeletal control.
- **App. F: Detailed Reward Curves for Reward-Based RL Baselines.** Presents the complete reward curves for all 22 benchmark tasks, enabling a detailed comparison of convergence behavior and training stability across different RL baselines.
- **App. G: Detailed Muscle Activation Analysis.** Provides time-step-resolved muscle activation heatmaps and biomechanical analysis for the 416-muscle and 700-muscle models, highlighting how anatomical fidelity affects load distribution and recruitment strategies.
- **App. H: Agentic Reward Tuning.** Describes the proposed closed-loop reward refinement mechanism, where an external reasoning module periodically adjusts reward weights according to recent gait quality, posture stability, and task-progress statistics.
- **App. I: Latent-Action RL.** Explains the expert-informed hard-mode action filtering framework, where a pretrained anatomical encoder-decoder constrains RL actions within a bounded neighborhood of biologically plausible muscle activations.
- **App. J: EMG Data Processing and Comparative Analysis.** Details the gait-cycle normalization, multi-cycle averaging, phase alignment, and Pearson-correlation-based comparison between simulated muscle activations and human EMG envelopes.
- **App. K: Muscle Recruitment Categorization.** Summarizes the anatomical keyword-based grouping of 416 muscle actuators into interpretable muscle families, allowing whole-body activation patterns to be analyzed at the functional group level.
- **App. L: Quantitative Load Intervention.** Reports

load-dependent regional activation shares and ratios under comparable task success.

- **App. M: Detailed Descriptions of Residual RL Policy.** Describes the residual reinforcement learning framework used to adapt the base tracking policy to challenging locomotion settings. This section details the residual action formulation, residual scaling strategy, Z-axis-agnostic imitation objective for stair navigation, elastic trajectory pacing, phase-dependent residual authority, metabolic-aware muscle regularization, and high-impact locomotion adaptations for running.
- **App. N: Limitations and Future Directions.** Discusses the current limitations of the benchmark and outlines future research directions, including dexterous manipulation, compositional and long-horizon task learning, broader physiological validation with EMG data, and sim-to-real transfer for physical musculoskeletal humanoid systems.
- **App. O: Extended Context and Interpretation.** Restores modeling background and clarifies comparison scope and residual/physiological interpretation.

A. Task Specification

Before enumerating the environment details below, let us define common auxiliary variables, symbols, and regularizations that are broadly employed across the reward functions of multiple environments. The complete task suite is visually summarized in Figs. 8–10, where Fig. 8 presents whole-body stabilization tasks, Fig. 9 illustrates representative locomotion behaviors and Fig. 10 shows terrain and object interaction tasks.

Kinematics and Dynamics:

- $q, \dot{q}$ denote the proprioceptive joint positions and velocities of the robot. Specifically, q_{joints} denotes the subset of non-root joint angles, and $\bar{q}_{\text{hip}}, \bar{q}_{\text{knee}}$ represent the average angles of the left and right hip/knee joints.
- $\mathbf{p}_{(\cdot)}$ represents the 3D position of a specific body part or object in the global frame (e.g., $\mathbf{p}_{\text{pelvis}}, \mathbf{p}_{\text{foot,L}}$). The superscript xy (e.g., $\mathbf{p}_{\text{pelvis}}^{xy}$) denotes its 2D horizontal projection.
- $\mathbf{v}_{(\cdot)}$ represents the 3D linear velocity vector. Specifically, $\mathbf{v}_{\text{CoM}}$ is the center-of-mass velocity, with $v_{x,\text{CoM}}, v_{y,\text{CoM}}, v_{z,\text{CoM}}$ being its scalar components.
- $z_{(\cdot)}$ denotes the vertical height of a specific body part or object from the ground (e.g., $z_{\text{CoM}}, z_{\text{pelvis}}, z_{\text{torso}}, z_{\text{head}}, z_{\text{foot}}$).
- $\Delta \mathbf{p}_{\text{feet}} = \mathbf{p}_{\text{foot,L}} - \mathbf{p}_{\text{foot,R}}$ denotes the relative 3D vector between the left and right feet. We use $d_{xy,\text{feet}}$ to denote the horizontal Euclidean separation between the feet, and $d_{x,\text{feet}}, d_{y,\text{feet}}, d_{z,\text{feet}}$ for its absolute scalar components.
- $\Delta z_{\text{leg}} = z_{\text{pelvis}} - \min(z_{\text{foot,L}}, z_{\text{foot,R}})$ defines the robot’s vertical leg extension, primarily used to enforce an upright posture and prevent deep-squatting or kneeling exploits.

Orientations and Vectors:

TABLE V

PROTOCOL-LEVEL COMPARISON COMPLEMENTING TAB. I. THIS COMPARISON CONCERNS A SHARED EVALUATION SET, RATHER THAN WHETHER A PLATFORM CAN BE EXTENDED TO SUPPORT A METRIC.

Dimension	MyoSuite	MyoChallenge	MSK-Bench
Task scope	Modular task environments	Annual competition tracks	22 full-body tasks
Task hierarchy	Task-specific variants	Track-specific progression	6 stabilization, 6 locomotion, 10 interaction
Embodiment	Model-specific	Track-specific	Shared 416-muscle full-body model
Evaluation	Task-specific	Track-specific	Seven common metric families
Physiology	No shared suite-wide set	Track-specific	Activation energy, smoothness, EMG where available
Methods	Method-agnostic platform	Competition submissions	Five paradigms within one protocol
Primary use	MSK simulation platform	Controller competition	Cross-task method evaluation

- $\mathbf{u}_{\text{torso}}$ is the normalized 3D unit vector pointing from the robot’s pelvis to its head, used to evaluate general core verticality.
- $\mathbf{u}_{\text{chest}}$ is the normalized unit vector pointing forward from the robot’s chest, used to penalize hunching or twisting.
- $\mathbf{u}_{\text{fwd}}$ is the normalized forward-facing vector derived from the pelvis orientation.
- $\hat{\mathbf{x}} = [1, 0, 0]^\top, \hat{\mathbf{y}} = [0, 1, 0]^\top, \hat{\mathbf{z}} = [0, 0, 1]^\top$ are the global forward, lateral, and vertical unit vectors.
- $y_{\text{yaw,dev}}$ denotes the forward yaw angular deviation derived from the robot’s pelvis orientation, utilized to restrict unintended turning.

Control, Environment, and Regularization:

- $a \in \mathbb{R}^M$ denotes the muscle activations at the current control step, with a_t and a_{t-1} specifying temporal steps.
- ϕ is the time-varying gait phase clock, represented in observations as $(\sin \phi, \cos \phi)$, with ω being its angular frequency.
- $\mathbf{z}_{\text{terrain}}$ is the local terrain observation grid representing relative ground heights.
- $\Delta \mathbf{h} = [\Delta x, h_{\text{target}}, 0]^\top$ denotes the relative 3D vector to an upcoming obstacle/hurdle.
- $\mathbb{I}_{\text{alive}}$ represents a constant survival bonus awarded at each step.
- $R_{\text{efficiency}} = \text{mean}(a^2)$ is the standard action regularization term applied across tasks to penalize excessive muscle force and promote energy-efficient behaviors. $\Sigma_{\text{penalties}}$ accumulates minor supplementary regularizations (e.g., joint limits).

The detailed reward functions, task settings, specific observations, and termination criteria across all 22 tasks in the Musculoskeletal Humanoid Benchmark are systematically summarized in Tables VI to VIII (Reward Functions) and Tables IX to XI (Setup and Termination).

B. Evaluation Details

In this section, we present a detailed quantitative evaluation of the benchmarked tasks. We systematically assess the performance, sample efficiency, and zero-shot robustness of various reinforcement learning baselines under increasingly difficult conditions. The detailed results for the Stabilization

(Table XII), Locomotion (Table XIII), and Interaction (Table XIV) series are summarized in the tables below.

1) *Training-Environment Success Rate:* The SR columns in the detailed tables evaluate the restored training environment, including the noise present during training. We denote this quantity by $\text{SR}_{\text{train-env}}$ in the main text. It should not be read as a noise-free reference value. The success event is defined below; restoring the training environment specifies the evaluation conditions. The fraction of test episodes in which the agent fulfills the task-specific success condition without falling or violating predetermined physical soft-constraints is defined as the success rate [1–3]. Tables display this fraction multiplied by 100 as a percentage. Higher success rate indicates better task performance. Formally, for N evaluation episodes, the success rate is computed as:

$$\text{Success Rate} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\mathcal{E}_{\text{task}}^{(i)} \wedge \mathcal{E}_{\text{alive}}^{(i)} \wedge \mathcal{E}_{\text{safe}}^{(i)}), \quad (1)$$

where $\mathbb{I}(\cdot)$ is the indicator function, $\mathcal{E}_{\text{task}}^{(i)}$ indicates whether the i -th episode meets the task-specific completion criterion, $\mathcal{E}_{\text{alive}}^{(i)}$ indicates whether the agent avoids falling, and $\mathcal{E}_{\text{safe}}^{(i)}$ indicates whether the agent stays within the predetermined physical soft-constraints.

For instance, in the walking task, an episode is deemed successful only if the agent survives for at least T_{min} and travels farther than a minimal goal distance d_{target} :

$$\text{Success Rate}_{\text{walk}} = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left(\left\| \mathbf{p}_{\text{end}}^{(i)} - \mathbf{p}_{\text{start}}^{(i)} \right\|_2 \geq d_{\text{target}} \wedge T_{\text{steps}}^{(i)} \geq T_{\text{min}} \right), \quad (2)$$

where N is the total number of evaluation episodes, $\mathbb{I}(\cdot)$ is the indicator function, $\mathbf{p}_{\text{start}}^{(i)}$ and $\mathbf{p}_{\text{end}}^{(i)}$ represent the agent’s starting and final planar pelvis or center-of-mass positions in the i -th episode, and $T_{\text{steps}}^{(i)}$ denotes the number of survived simulation steps.

2) *Perturbation-Sweep Robustness:* The action, observation, and dynamics columns report separate perturbation sweeps. Their zero-axis entries and native noisy training-environment SR are distinct reports; equality is not an

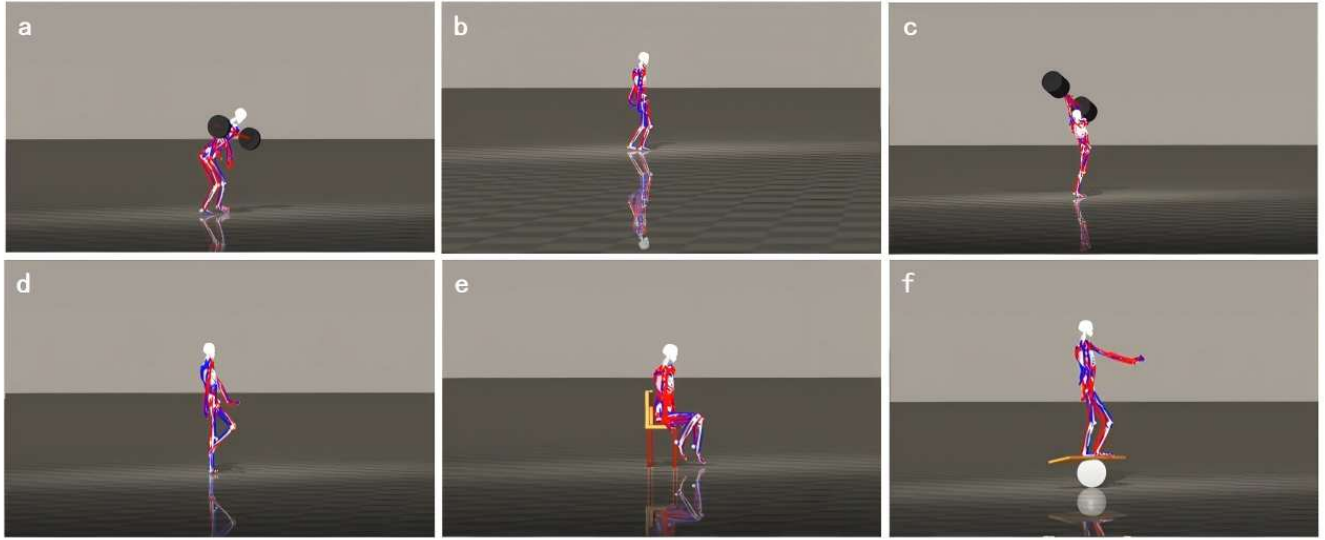


Fig. 8. **Illustration of six representative whole-body stabilization tasks.** (a) **squat**: The control policy must accurately track the dynamic vertical trajectories of the hip and knee based on a phase clock ϕ , while maintaining a strict center-of-mass balance lock between the two feet. A specially designed “landmine penalty” is introduced to immediately eliminate abnormal strategies that attempt to exploit the height-tracking reward by kneeling on both knees. (b) **stand**: The robot is required to maintain an absolutely static upright posture from an initial neutral pose. The key challenge is that the system must use high muscle stiffness to lock the joints, while even slight center-of-mass drift or foot sliding is severely penalized. The reward function is strongly biased toward absolute stillness and posture maintenance. (c) **powerlift**: To sustain the extreme load, the reward function explicitly encourages a biomechanically plausible forward-leaning hip-hinge posture through the R_{hinge} reward, thereby effectively recruiting the powerful posterior-chain muscle groups. Once dangerous lumbar collapse is detected, the episode is immediately terminated to ensure structural safety. (d) **singlelegstand**: The robot must lift one leg into a “flamingo” posture and strictly align the center-of-mass projection with the support foot. The task severely penalizes lateral trunk lean, and after an initial physical tolerance period of 10 steps, the episode is immediately terminated once the lifted foot touches the ground. (e) **sit**: The robot is required to approach a chair and execute a precise sitting transition. To prevent cheating behaviors such as stiff-legged falling, the task enforces active knee flexion, requires the feet to be retracted beneath the support area, and introduces a “soft-landing” reward to limit pelvic descent velocity, ensuring a smooth motion. (f) **balance**: The robot is required to maintain dynamic balance on an unconstrained wobble board placed on a sphere. To overcome the common failure mode of falling backward, the task explicitly enforces a forward-leaning “fighting defensive stance”, with a target pitch angle of -0.15rad , and requires precise center-of-mass control to keep the balance board level.

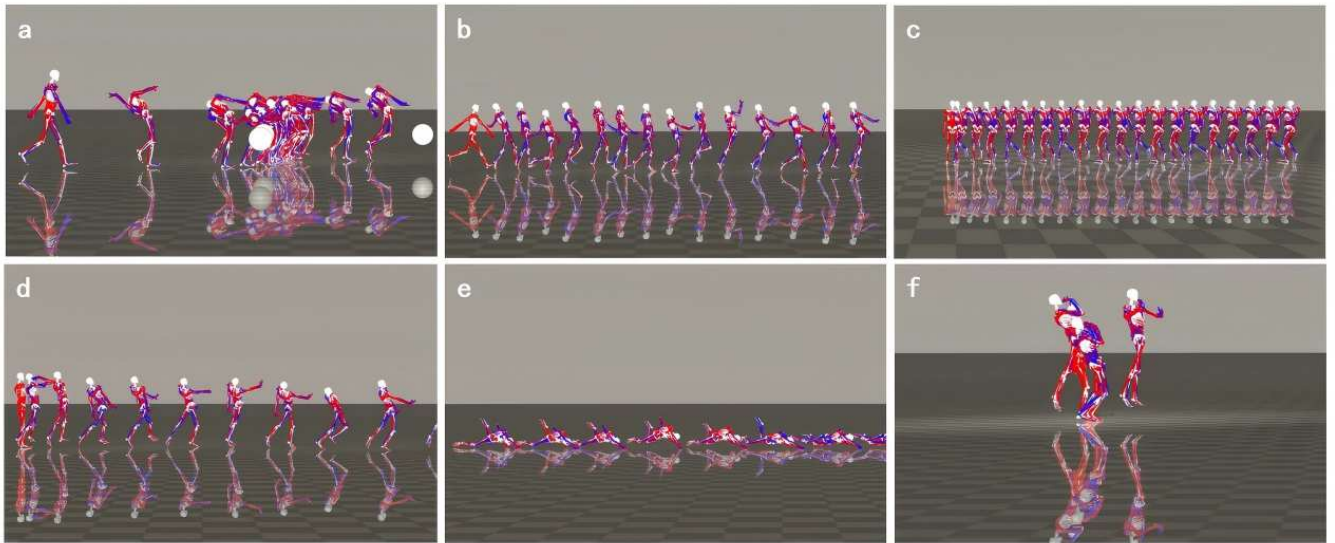


Fig. 9. **Visualization of representative locomotion behaviors generated by the humanoid robot.** Each subfigure shows a sequence of motion frames, with red and blue skeletons indicating the two legs and body segments over time. (a) **walk turn**: The robot moves while interacting with spherical obstacles. (b) **walk forward**: The robot demonstrates a stable periodic gait with alternating leg support. (c) **sidestep**: The robot demonstrates sustained rhythmic whole-body coordination. (d) **run**: The robot maintains balance while moving on a wheeled board. (e) **crawl**: The robot performs low-posture locomotion with the body close to the floor. (f) **jump**: The robot performs a dynamic aerial transition while maintaining body balance.

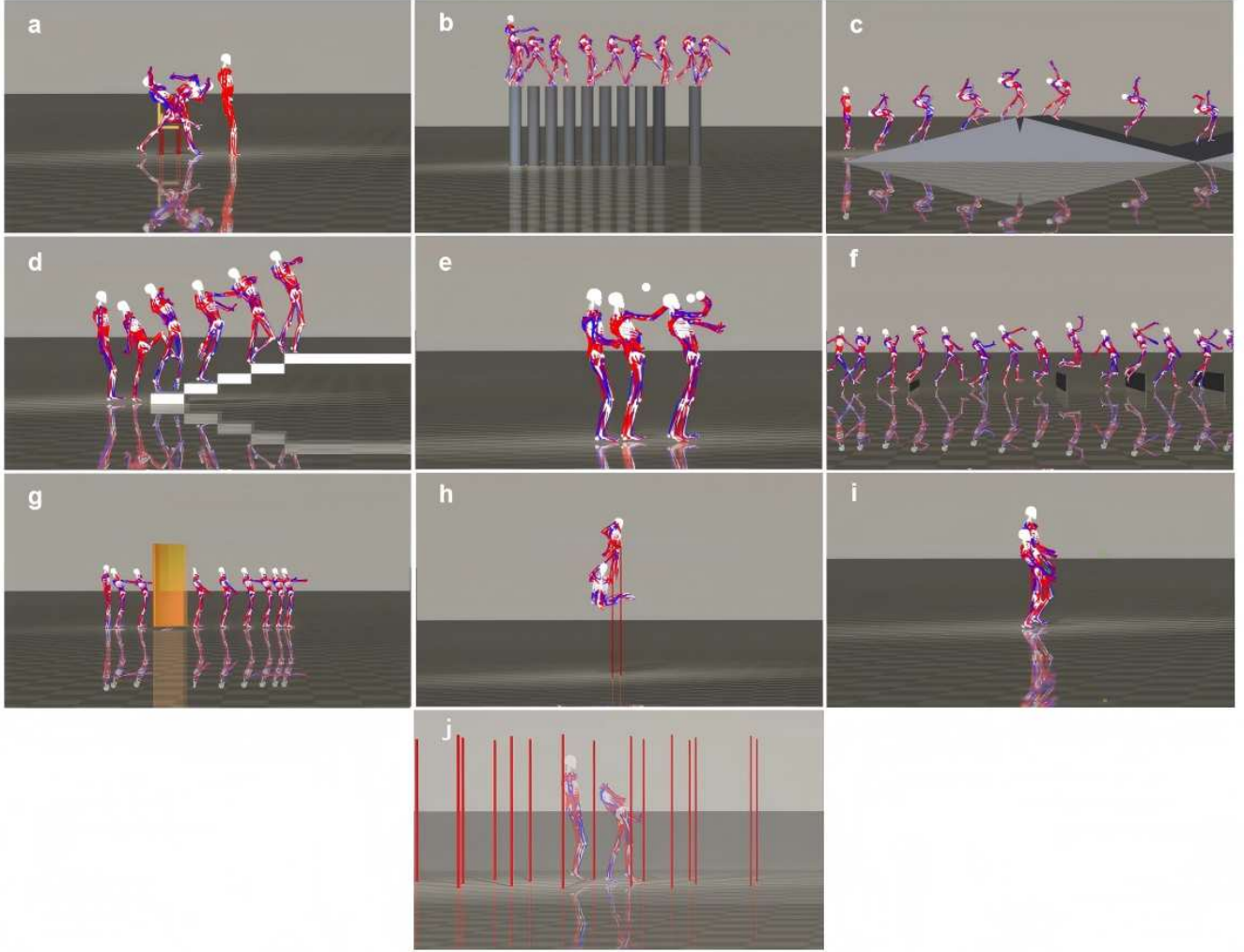


Fig. 10. **Illustration of ten representative terrain and object interaction tasks.** (a) **walkandsit**: The robot is required to approach a chair backward and sit down. Instead of using a conventional collision-avoidance penalty, the task rewards the robot for confidently utilizing downward inertia to complete the sitting motion and for comfortably leaning back against the chair after sitting, thereby reproducing human-like sitting biomechanics. (b) **stepstones**: A magnetic attraction reward is used to guide the foot toward the target landing positions, while strict penalties on trunk lateral bending, yaw deviation, and excessive knee flexion encourage an upright and stable traversal posture. (c) **slide**: The robot is required to adaptively lean forward to counteract inertia during slope transitions. A split-prevention penalty is introduced to terminate the episode once the lateral separation of the legs exceeds the biomechanical limit. (d) **stairs**: The robot is required to climb stairs using look-ahead terrain perception from a terrain radar. When upcoming steps are detected, an active leg-lifting reward is activated to encourage anticipatory step planning. (e) **reach**: The robot is required to rapidly and explosively reach randomly reset spatial targets with its dominant hand while maintaining a slightly crouched defensive posture. Since the target height varies, the task strictly constrains knee height to prevent cheating strategies such as kneeling. (f) **hurdle**: The task enforces a complete aerial flight phase and introduces a strict anti-flamingo-jump penalty to prevent abnormal strategies in which the robot crosses the hurdle by curling and dragging a single leg. (g) **open door**: The robot is required to approach, push open, and pass through a hinged door. A global attraction field guides the hand toward the door handle, while fall-prevention penalties discourage shortcut behaviors such as headbutting the door or diving through it. (h) **chinup**: The robot is required to pull its head above the horizontal bar from a hanging posture. The task rewards the activation of major pulling muscle groups such as the latissimus dorsi and biceps, while imposing strict penalties on body swinging and leg-kipping, forcing the motion to be completed through pure muscular strength. (i) **catch**: The robot is required to coordinate both arms to catch a flying cube thrown toward the chest while maintaining an upright posture. The key objective is to track the distance between both hands and the target, and a large sparse reward is given when the object is successfully caught and stabilized in mid-air. (j) **polewalk**: The robot is required to scan surrounding poles using terrain radar and traverse the entire pole region without collision. The episode is immediately terminated once any body part touches a pole.

TABLE VI

REWARD FUNCTIONS FOR ALL 22 MUSCULOSKELETAL BENCHMARK TASKS (PART 1: STABILIZATION SERIES). NOTE: ACTION REGULARIZATION ($R_{\text{EFFICIENCY}}$) AND MINOR LIMIT PENALTIES (Σ PENALTIES) ARE APPLIED BROADLY ACROSS TASKS AND ABBREVIATED WHERE STANDARD.

Task	Term	Formula	Weight
1. Stand	Pose preservation	$\exp(-\ q_{\text{joints}} - q_{0,\text{joints}}\)$	10
	Absolute stillness	$\exp(-10.0\ \mathbf{v}_{\text{CoM}}\) \cdot \exp(-0.1\ \dot{q}\)$	15
	Feet planted	$\exp(-10.0(\ \mathbf{v}_{\text{foot,L}}\ + \ \mathbf{v}_{\text{foot,R}}\))$	10
	Torso balance	$\exp(-5.0\max(0, 1 - \mathbf{u}_{\text{torso}} \cdot \hat{\mathbf{z}}))$	10
	Survival	$\mathbb{I}_{\text{alive}}$	5
2. Powerlift	Hold stability	$\mathbb{I}(z_{\text{bar}} > 1.6\text{m} \wedge \dot{z}_{\text{bar}} < 0.1\text{m/s})$	100
	Height threshold	$\mathbb{I}(z_{\text{bar}} > 1.6\text{m})$	50
	Upright alignment	$\exp(-20.0(1 - \mathbf{u}_{\text{torso}} \cdot \hat{\mathbf{z}}))$	100
	Lumbar spine safety	$\exp(-15.0 q_{\text{lumbar}} - (-0.4))$	80
	Muscular effort	$\text{mean}(a)$	30
	Hip hinge posture	$\mathbb{I}(q_{\text{lumbar}} > 0.1)$	200
3. Singleleg	Balance (CoM)	$\exp(-15.0\ \mathbf{p}_{\text{pelvis}}^{\text{xy}} - \mathbf{p}_{\text{foot,R}}^{\text{xy}}\ ^2)$	500
	Lifted foot clear	$\text{clip}(z_{\text{foot,L}}/0.3, 0, 1)$	300
	Absolute stillness	$\exp(-5.0\ \mathbf{v}_{\text{CoM}}\ ^2)$	400
	Upright alignment	$\exp(-10.0(1 - \mathbf{u}_{\text{torso}} \cdot \hat{\mathbf{z}})^2)$	200
	Foot touch penalty	$\mathbb{I}(z_{\text{foot,L}} < 0.05\text{m})$	-800
	Lateral lean penalty	$ \mathbf{u}_{\text{torso}} \cdot \hat{\mathbf{y}} $	-200
4. Sit	Chair approach	$2.0/(\ \Delta\mathbf{p}_{\text{chair}}^{\text{xy}}\ + 0.2)$	10
	3D Contact docking	$\exp(-\frac{\Delta x^2}{2 \cdot 0.15^2}) \cdot \exp(-\frac{\Delta z^2}{2 \cdot 0.15^2})$	30
	Lateral centering	$\exp(-\frac{\Delta y^2}{2 \cdot 0.05^2})$	30
	Seated leg geometry	$\exp(-2.0 \bar{q}_{\text{hip}} - 1.5 - k_{\text{knee}} \bar{q}_{\text{knee}} - 1.6)$	40
	Feet tucked under	$\exp(-5.0\sum_i (x_{\text{foot},i} - x_{\text{pelvis}}) - 0.2)$	30
	Soft landing	$\exp(-2.0 \dot{z}_{\text{pelvis}})$	5
5. Balance	Forward lean stance	$\exp(-10.0(\theta_{\text{pitch}} - (-0.15))^2)$ (gated)	15
	CoM over board	$\exp(-20.0\ \mathbf{p}_{\text{CoM}}^{\text{xy}} - \mathbf{p}_{\text{board}}^{\text{xy}}\ ^2)$	15
	Board over ball	$\exp(-5.0\ \mathbf{p}_{\text{board}}^{\text{xy}} - \mathbf{p}_{\text{ball}}^{\text{xy}}\)$	5
	Board level	$\exp(-5.0(1 - \mathbf{u}_{\text{board}} \cdot \hat{\mathbf{z}}))$	5
	Stand tall	$\exp(-10.0(1.30 - z_{\text{pelvis}})^2)$ if $z < 1.3$	4
6. Squat	Height tracking	$\exp(-30.0(z_{\text{pelvis}} - z_{\text{target}})^2)$	200
	Joint tracking	$\exp(-5.0(\bar{q} - q_{\text{target}})^2)$	300/200
	Vertical vel tracking	$\exp(-5.0(v_{z,\text{CoM}} - v_{z,\text{target}})^2)$	200
	CoM centered	$\exp(-15.0\ \mathbf{p}_{\text{pelvis}}^{\text{xy}} - \mathbf{p}_{\text{mid,feet}}^{\text{xy}}\ ^2)$	300
	Kneeling penalty	$\mathbb{I}(\min(z_{\text{knee,L}}, z_{\text{knee,R}}) < 0.25\text{m})$	-1000

evaluation requirement. Action noise, observation noise, and dynamics randomization are introduced to assess robustness [4, 5] of the methods. It evaluates whether a policy can maintain balance, preserve task performance, or quickly recover its intended trajectory when facing stochastic perturbations and randomized dynamics [4, 5]. To comprehensively test robustness under Sim2Real-style perturbations [6–9], we consider the following:

First, **action noise** mimics stochastic execution errors in motor-neural excitation signals or low-level actuator commands. Sim-to-real robot learning studies that model stochastic action effects and transition mismatch between simulation and the real world [10], together with continuous-control reinforcement learning studies that examine additive action noise [11], motivate such action-space perturbations. In our implementation, Gaussian noise is applied to the normalized action vector generated by the policy. Since the policy action space is bounded within $[-1, 1]$, the perturbed action is

clipped to this valid action range:

$$\begin{aligned} \mathbf{u}'_t &= \text{clip}_{[-1,1]}(\mathbf{u}_t + \boldsymbol{\epsilon}_t^{\text{act}}), \\ \boldsymbol{\epsilon}_t^{\text{act}} &\sim \mathcal{N}(\mathbf{0}, \sigma_{\text{act}}^2 \mathbf{I}), \end{aligned} \quad (3)$$

where $\mathbf{u}_t$ denotes the policy’s original normalized action output, $\boldsymbol{\epsilon}_t^{\text{act}}$ denotes Gaussian action noise, σ_{act} controls the action noise magnitude, and $\mathbf{u}'_t$ denotes the noisy action applied to the environment.

Second, **observation noise** mimics sensing errors from external sensors such as IMUs or proprioceptive feedback such as muscle spindles and Golgi tendon organs. This perturbation is also consistent with prior sim-to-real studies that perturb or randomize sensory inputs to improve robustness under observation mismatch [6, 7, 9]. In our implementation, Gaussian noise is added to the normalized policy observation

TABLE VII
REWARD FUNCTIONS FOR MUSCULOSKELETAL BENCHMARK TASKS (PART 2: LOCOMOTION SERIES).

Task	Term	Formula	Weight
1. Walk	Velocity tracking	$\exp(-2.0\ \mathbf{v}_{\text{CoM}} - \mathbf{v}_{\text{target}}\ ^2)$	10
	Kinematic imitation	$\exp(-2.0\ q_{\text{joints}} - q_{\text{ref,joints}}\ ^2)$	3
	Upright / No hunch	$\exp(-10.0(1 - \mathbf{u}_{\text{torso}} \cdot \hat{\mathbf{z}})), \exp(-20.0(\mathbf{u}_{\text{chest}} \cdot \hat{\mathbf{z}} + 0.1)^2)$	6/8
	Forward face	$\exp(-5.0(1 - \mathbf{u}_{\text{chest}} \cdot \hat{\mathbf{x}}))$	8
2. Crawl	Backwards velocity	$\text{clip}(-v_{x,\text{CoM}}/v_{\text{target}}, -0.2, 1.0)$	500
	Prone posture	$\exp(-5(\mathbf{u}_{\text{chest}} \cdot \hat{\mathbf{z}} + 1)^2) \cdot \dots$	100
	Alternating stride	$\text{clip}(\Delta x_{\text{hands}} /0.3, 0, 1) \cdot \dots$ (diag gait)	150
	Drift penalty	$2.0 v_{y,\text{CoM}} + v_{x,\text{CoM}} \mathbb{I}(v_x > 0)$	-150
3. Run	High velocity	$\text{clip}(v_{x,\text{CoM}}, -0.5, 6.0)$, bonus if $v > 3.0$	200/100
	Asymmetric stride	$\text{clip}(d_{x,\text{feet}} + 2.0 \cdot d_{z,\text{feet}}, 0.0, 1.0)$	200
	Knee flexion (swing)	$\text{clip}((\max(q_{\text{knee}}) - 0.1)/1.0, 0, 1)$	200
	Scissor penalty	$\text{clip}((0.15 - d_{y,\text{feet}})/0.15, 0, 1)$	-300
	Kangaroo penalty	Penalize $d_{x,\text{feet}} < 0.2 \wedge d_{z,\text{feet}} < 0.1$	-300
4. Jump	Flight gradient	$\text{clip}(\frac{1}{2}(z_{\text{foot,L}} + z_{\text{foot,R}})/0.1, 0, 1)$	600
	Upward explosion	$\text{clip}(\dot{z}_{\text{pelvis}}, 0, 2.0)$	300
	Feet separation pen.	$d_{xy,\text{feet}}$	-200
	Desync jump pen.	$ q_{\text{knee,L}} - q_{\text{knee,R}} + q_{\text{hip,L}} - q_{\text{hip,R}} $	-150
5. Walk turn	Targeted velocity	$\text{clip}(\mathbf{v}_{\text{CoM}} \cdot \mathbf{u}_{\text{target}}/3.0, -1.0, 1.0)$	200
	Waypoint approach	$\exp(-\ \Delta \mathbf{p}_{\text{target}}\)$	50
	Target facing	$\text{clip}(\mathbf{u}_{\text{fwd}} \cdot \mathbf{u}_{\text{target}}, 0, 1)$	50
	Waypoint reached	$\mathbb{I}(\ \Delta \mathbf{p}_{\text{target}}\ < 0.5\text{m})$	500
	Wall collision	$\mathbb{I}_{\text{contact}}(\text{wall})$	-200
6. Sidestep	Lateral velocity	$\text{clip}(v_{y,\text{CoM}}/v_{\text{target}}, -0.2, 1.0)$	400
	Phase sync	$\exp(-15.0(e_L + e_R))$	200
	CoM mid-foot lock	$ y_{\text{pelvis}} - \frac{1}{2}(y_{\text{foot,L}} + y_{\text{foot,R}}) $	-500
	Rubber band splits	$\max(0, d_{xy,\text{feet}} - 0.3)$	-400

vector before it is passed to the policy network:

$$\begin{aligned} \tilde{\mathbf{s}}'_t &= \text{clip}_{[-20,20]}(\tilde{\mathbf{s}}_t + \boldsymbol{\epsilon}_t^{\text{obs}}), \\ \boldsymbol{\epsilon}_t^{\text{obs}} &\sim \mathcal{N}(\mathbf{0}, \sigma_{\text{obs}}^2 \mathbf{I}), \end{aligned} \quad (4)$$

where $\tilde{\mathbf{s}}_t$ denotes the normalized observation vector before perturbation, $\boldsymbol{\epsilon}_t^{\text{obs}}$ denotes Gaussian observation noise, σ_{obs} controls the observation noise level, and $\tilde{\mathbf{s}}'_t$ denotes the perturbed normalized observation received by the policy. The clipping operation is applied element-wise to keep the noisy normalized observation within the predefined numerical range.

Third, **dynamics randomization** mimics individual variations in muscle physiology and uncertainty in muscle force capacity. To improve robustness against real-world dynamics mismatch [7–9], this setting follows the broader sim-to-real practice of randomizing physical parameters, such as mass, friction, actuator properties, and other dynamics-related quantities. At the beginning of each evaluation episode, we independently sample a random perturbation for each muscle and keep it fixed throughout the episode. Specifically, for the m -th muscle, we sample:

$$\begin{aligned} \delta_m &\sim \mathcal{U}(-\sigma_{\text{dyn}}, \sigma_{\text{dyn}}), \\ r_m &= 1 + \delta_m, \\ \text{Gain}'_m &= \text{Gain}_m \times r_m, \end{aligned} \quad (5)$$

where δ_m denotes the independently sampled perturbation for the m -th muscle, r_m denotes the corresponding multiplicative gain factor, Gain_m denotes the original maximum force-generation parameter of the m -th muscle, and Gain'_m denotes the randomized muscle gain used during evaluation. The parameter $0 \leq \sigma_{\text{dyn}} < 1$ controls the perturbation strength and ensures positive muscle gains. For example, $\sigma_{\text{dyn}} = 0.20$ means that each muscle’s maximum force capacity is independently scaled by a random factor in $[0.8, 1.2]$.

Across these perturbation settings, we report the average reward, survival steps, and true success rate to quantify how reliably each policy maintains task performance under noise and randomized dynamics. Higher values of these metrics under perturbation indicate stronger robustness. Within each sweep, performance changes are interpreted relative to that sweep’s zero-axis entry, not relative to the separately reported training-environment SR.

3) *Muscle Activation Energy*: Muscle activation energy [4, 12–15] is used to quantify the average muscle activation cost by computing the average squared muscle activation of all underlying muscles across the whole body within a time window. This metric is used as a proxy for muscle effort and physiological efficiency of the generated motion. Since the human central nervous system tends to optimize energy consumption, this metric helps filter out poor-quality policies that maintain balance by relying on excessive muscle

TABLE VIII
REWARD FUNCTIONS FOR MUSCULOSKELETAL BENCHMARK TASKS (PART 3: INTERACTION WITH ENVIRONMENT SERIES).

Task	Term	Formula	Weight
1. Stairs	Fwd trajectory	$\text{clip}(v_{x,\text{CoM}}, -0.2, 2.0)$	50
	Upward trajectory	$\text{clip}(v_{z,\text{CoM}}, -0.5, 3.0)$	30
	Vertical stable	$\exp(-5.0(0.98 - \mathbf{u}_{\text{torso}} \cdot \hat{\mathbf{z}})^2)$	20
	Lateral stable	$\exp(-5.0y_{\text{pelvis}}^2)$	20
	Anticipatory lift	$\mathbb{I}(\text{radar}) \cdot \max(\dot{z}_{\text{swing}}, 0)$	40
2. Hurdle	Forward velocity	$\text{clip}(v_{x,\text{CoM}}, -0.1, 4.0)$	300
	Clearance (Flight)	$\mathbb{I}(\max(z_{\text{feet}}) > h_{\text{target}})$	100
	Sparse clear bonus	$\mathbb{I}(\Delta x < 0 \wedge y_{\text{pelvis}} < 0.5\text{m})$	3000
	Crane hopping pen.	$\text{clip}((z_{\text{foot,L}} - z_{\text{foot,R}} - 0.2)/0.5, 0, 1)$	-150
3. Step stones	Target stone progress	Bonus when crossing stone X-coord	800
	Magnetic reaching	$\exp(-2.0 \cdot d_{\text{lead}}) \cdot \mathbb{I}(v_{x,\text{CoM}} > 0.1)$	150
	Stand tall	$\text{clip}((\Delta z_{\text{leg}} - 0.6)/0.3, 0, 1)$	50
	Overbend knee pen.	$\max(q_{\text{knee}} - 1.0, 0)$	-50
4. Slide	Pitch balance	$R_{\text{gate}} \cdot \exp(-5.0 \cdot \theta_{\text{pitch}}^2)$	2
	Stand tall	$R_{\text{gate}} \cdot \exp(-10.0(0.95 - \Delta z_{\text{leg}})^2)$	2
	Slope forward lean	$\text{clip}(-\mathbf{u}_{\text{chest}} \cdot \hat{\mathbf{z}}/0.3, 0, 1)$	20
	Backward lean pen.	$\max(\mathbf{u}_{\text{chest}} \cdot \hat{\mathbf{z}}, 0)$	-100
5. Open door	Hand attractor field	$\exp(-1.5\ \Delta \mathbf{p}_{\text{handle}}\)$	400
	Open hinge	$\text{clip}(\theta_{\text{door}}/1.2, 0, 1) \cdot \text{prox_gate}$	500
	Pass threshold	Sparse bonuses for $x > 1.0, 2.0$	2000
	Dive/Split penalties	Penalize pelvis lead or wide feet	-300/-400
6. Reach	Dense reach dist	$\exp(-4.0\ \Delta \mathbf{p}_{\text{reach}}\)$	50
	Target velocity proj	$\text{clip}(\mathbf{v}_{\text{hand}} \cdot \mathbf{u}_{\text{reach}}, 0, 2.0)$	20
	Target touch bonus	$\mathbb{I}(\ \Delta \mathbf{p}_{\text{reach}}\ < 0.15\text{m})$	200
7. Walk & sit	Backwards precision	$\exp(-\ \Delta \mathbf{p}_{\text{chair}}^{\text{xy}}\ ^2/0.005)$	100
	Fast descent bonus	$\mathbb{I}(\dot{z}_{\text{pelvis}} < -1.5\text{m/s})$	50
	Relaxed recline	$\mathbb{I}(\mathbf{u}_{\text{torso}} \cdot \hat{\mathbf{z}} < 0.8)$ (once seated)	200
	90-degree pose	$\exp(-5(\bar{q}_{\text{hip}} - 1.5)^2) + \dots$	100
8. Chin up	Spatial gradient	$\exp(-5.0 \max(0, h_{\text{bar}} - z_{\text{head}}))$	50
	Muscle force filter	$\text{mean}(F_{\text{pull_muscles}})$	5
	Explosive accel	$\max(0, \ddot{z}_{\text{pelvis}})$	40
	Anti-kipping wiggle	$\ \dot{q}_{\text{legs}}\ $	-20
9. Catch	Hand-object reach	$\exp(-3.0 \cdot d_{\text{catch}})$	100
	Interception bonus	$\mathbb{I}(d_{\text{catch}} < 0.15\text{m} \wedge z_{\text{cube}} > 0.5\text{m})$	500
	Drop penalty	$\mathbb{I}(z_{\text{cube}} < 0.1\text{m})$	-200
10. Pole walk	Target feet phases	$\exp(-10.0(\Delta x_{\text{foot}} \pm 0.25 \sin \phi)^2)$	400
	Dynamic radar gap	E.g., $\text{clip}(2.0 \cdot v_{y,\text{CoM}}, 0, 1)$ if left gap	800
	Fatal collision	$\mathbb{I}_{\text{fatal_contact}}$ (head/torso to pole)	-1500

TABLE IX

TASK SETTINGS, SPECIFIC OBSERVATIONS, AND TERMINATION CRITERIA (PART 1: STABILIZATION SERIES). *Base Observation (O_{base}): All tasks fundamentally include proprioceptive states ($q, \dot{q}, \mathbf{v}_{\text{CoM}}$) and muscle states (length, velocity, force, activation). Only task-specific exteroceptive or goal variables are listed below.*

Task	Specific Obs (O_{task})	Initialization (Brief)	Termination Criteria
1. Stand	Torso quat, feet height	Default neutral, zero vel.	$z_{\text{CoM}} < 0.8$ or tilt > 0.8
2. Powerlift	Dumbbell $\mathbf{p}_{\text{bar}}, \Delta \mathbf{p}_{\text{bar}}$, tilt	High CoM, pre-loaded lumbar	$z_{\text{pelvis}} < 0.7$ or $q_{\text{lumbar}} > 0.5$
3. Singleleg	Torso quat, feet/CoM z	Flamingo pose, noise	$z_{\text{pelvis}} < 0.6$, dropped foot
4. Sit	Chair relative $\Delta \mathbf{p}_{\text{chair}}$	In front of chair, hips pre-flexed	$z_{\text{pelvis}} < 0.25$, invalid contact
5. Balance	Board $\mathbf{p}, \mathbf{q}$, CoM/Ball $\Delta \mathbf{p}$	Drop $z = 1.28\text{m}$, guard stance	$z_{\text{pelvis}} < 0.70$, pitch fall
6. Squat	Phase ϕ , target z_{target}	Squat start pose, barbell hold	$z_{\text{pelvis}} < 0.35$, kneeling

TABLE X
TASK SETTINGS, SPECIFIC OBSERVATIONS, AND TERMINATION CRITERIA (PART 2: LOCOMOTION SERIES).

Task	Specific Obs (O_{task})	Initialization (Brief)	Termination Criteria
1. Walk	Phase ϕ , error $\Delta q = q - q_{\text{ref}}$	Sampled from synthetic walk	$z_{\text{CoM}} < 0.5$ or tilt > 0.8
2. Crawl	Torso quat, feet height	Prone quadruped, ground $z = 0.25$	Fwd move, twist, flip
3. Run	Torso angle, feet height	Standard standing	$z_{\text{pelvis}} < 0.45$, fall backward
4. Jump	Torso angle, $d_{\text{xy,feet}}$	Micro-squat, pre-flexed	$z_{\text{pelvis}} < 0.45$, wide split
5. Walk turn	Target $\Delta \mathbf{p}$, ID, facing a	Stand, 20-step novice protection	$z_{\text{pelvis}} < 0.55$, $z_{\text{head}} < 0.8$
6. Sidestep	Phase ϕ , lat dist Δy_{feet}	Crab guard stance, hips wide	$z_{\text{pelvis}} < 0.45$, split $> 0.5\text{m}$

TABLE XI
TASK SETTINGS, SPECIFIC OBSERVATIONS, AND TERMINATION CRITERIA (PART 3: INTERACTION WITH ENVIRONMENT SERIES).

Task	Specific Obs (O_{task})	Initialization (Brief)	Termination Criteria
1. Stairs	Phase ϕ , radar $\mathbf{z}_{\text{terrain}} \in \mathbb{R}^9$	Walk posture, initial fwd vel	$z_{\text{pelvis}} < 0.75$, lat drift
2. Hurdle	Phase ϕ , nearest $\Delta \mathbf{h}$	Standing, scan dynamic hurdles	$z_{\text{pelvis}} < 0.55$, heavy yaw
3. Step stones	Phase ϕ , stones $\Delta \mathbf{p}_{\text{stones}}$	Elevated platform $z = 3.05\text{m}$	Fall from height, $z_{\text{pelvis}} < 2.5$
4. Slide	Phase ϕ , normal $\mathbf{n}_{\text{slope}}$	Elevated over wave terrain	Backwards fall, split $> 1.2\text{m}$
5. Open door	Handle $\Delta \mathbf{p}$, door θ , target	Stand front of closed door	$z_{\text{pelvis}} < 0.65$, head < 1.1
6. Reach	Phase ϕ , reach $\Delta \mathbf{p}_{\text{reach}}$	Quarter-squat, hands raised	Knee drop $< 0.15\text{m}$
7. Walk & sit	Chair $\Delta \mathbf{p}$, phase $\mathbb{I}_{\text{seated}}$	Face away, inject backward vel	Ground contact (head/knees)
8. Chin up	Phase (0), bar dist Δz_{chin}	Hanging, flexed shoulders	Fall off bar ($z_{\text{pelvis}} < 0.5$)
9. Catch	Cube $\mathbf{p}, \mathbf{v}$, dist d_{catch}	Anticipatory basket, cube spawn	Drop cube ($z_{\text{cube}} < 0.05$)
10. Pole walk	Phase ϕ , radar $\mathbf{z}_{\text{radar}} \in \mathbb{R}^7$	Elevated 1.05m, high core gain	Fatal upper-body collision

activation or co-contraction [14, 15]. Formally, the mean squared muscle effort is computed as:

$$E_{\text{muscle}} = \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{M} \sum_{m=1}^M a_{m,t}^2 \right), \quad (6)$$

where $a_{m,t} \in [0, 1]$ denotes the actuator activation of the m -th muscle at time step t , T denotes the total number of simulation steps in each episode, and M denotes the total number of muscles. This is an activation-based effort proxy, not calibrated metabolic energy. Lower values are favorable only when task performance and evaluated behavior are comparable. For example, in walking SAC has cost 0.1427 but 0% SR, whereas DepRL has cost 0.4206 and 64% SR. Ranking these costs alone would reward failure. We therefore interpret activation cost and jerk alongside success, rather than declare an overall physiological winner from unconditioned costs.

4) *Joint Smoothness*: Joint smoothness is evaluated by analyzing high-frequency fluctuations in joint motion. Prior work has used smoothness-related kinematic measures to characterize coordinated and biologically plausible movement [16, 17]. Lower high-frequency jerk indicates smoother motor behavior with fewer abrupt joint-level changes, which is consistent with the low-pass characteristics of biological motor control [16, 17].

In our implementation, smoothness is computed from the joint angular velocity signal returned by the physics engine, denoted as $v_{j,t}$ for the j -th joint at time step t . Since our implementation applies a second-order finite difference to joint velocity, the resulting finite-difference quantity is

proportional to the third-order derivative of joint angle, commonly referred to as jerk:

$$\begin{aligned} D^{(2)}v_{j,t} &= \frac{v_{j,t} - 2v_{j,t-1} + v_{j,t-2}}{\Delta t^2} \\ &\approx \frac{d^2 v_j(t)}{dt^2} = \frac{d^3 \theta_j(t)}{dt^3}. \end{aligned} \quad (7)$$

We therefore quantify joint-level high-frequency fluctuation using the mean squared finite-difference jerk over all joints and time steps:

$$J_{\text{sq}} = \frac{1}{(N-2)N_{\text{joint}}} \sum_{t=3}^N \sum_{j=1}^{N_{\text{joint}}} \left(D^{(2)}v_{j,t} \right)^2, \quad (8)$$

where N_{joint} denotes the total number of joints, N denotes the total number of simulation steps in the episode, $v_{j,t}$ denotes the joint angular velocity of the j -th joint at time step t , Δt denotes the simulation time step, and $D^{(2)}v_{j,t}$ denotes the second-order finite difference of joint angular velocity.

For visualization and task-level comparison, we report the logarithmic mean squared jerk:

$$\text{Smoothness}_{\text{jerk}} = \log_{10} (J_{\text{sq}} + \epsilon), \quad (9)$$

where ϵ is a small constant used for numerical stability. Lower values indicate smoother motion with fewer high-frequency joint fluctuations, whereas higher values indicate less smooth and more abrupt joint motion.

5) *Peak Efficiency Steps*: We introduce this metric to identify the training phase at which an algorithm achieves its peak evaluation performance. Peak Efficiency Steps pinpoints the number of environment interactions required to

obtain the highest expected evaluation return over the full training horizon. Formally, it is defined as the evaluated training step t that maximizes the expected evaluation return:

$$T_{\max} = \arg \max_{t \in \mathcal{T}} \mathbb{E}[R_t], \quad (10)$$

where $\mathcal{T}$ denotes the set of evaluated training steps throughout the learning process, and R_t denotes the cumulative reward obtained during evaluation at training step t . We also denote the corresponding maximum expected return as

$$R_{\max} = \max_{t \in \mathcal{T}} \mathbb{E}[R_t]. \quad (11)$$

A smaller $T_{\max}$, particularly when paired with a high $R_{\max}$, indicates that the algorithm can rapidly discover a high-performing behavioral policy, reflecting better sample efficiency and faster peak skill acquisition.

6) *EMG Similarity*: EMG similarity compares simulated muscle activation envelopes with human recordings for walking and stairs from Gait120 [18], and running at 3 m/s from Van Hooren and Meijer [19], following the activation-envelope interpretation used in MuscleMimic [20]. The extended running/stairs analysis uses 12 matched lower-limb muscles (Tab. IV). Availability of matched recordings determines coverage; EMG is not reported for every task. Since multiple muscle coordination strategies can generate similar joint kinematics, and accurate motion tracking does not necessarily imply human-like muscle recruitment, we treat the simulated muscle activation state as an activation-envelope proxy [15, 20, 21].

For each selected lower-limb muscle, simulated activations and human EMG recordings are processed in a gait-cycle-normalized form. Human EMG envelopes from the corresponding datasets [18, 19] are rectified, normalized on a per-muscle basis, temporally aligned to the corresponding movement cycle, and resampled to a common normalized stride representation [18, 20]. Muscle activations from RL are segmented into gait cycles, interpolated to the same normalized time base, and averaged across cycles to obtain representative activation profiles. We then compare the waveform similarity between the simulated activation envelope and the processed human EMG envelope using Pearson correlation:

$$r_m = \frac{\sum_{t=1}^T (a_{\text{sim},m}(t) - \bar{a}_{\text{sim},m})(e_{\text{EMG},m}(t) - \bar{e}_{\text{EMG},m})}{\sqrt{\sum_{t=1}^T (a_{\text{sim},m}(t) - \bar{a}_{\text{sim},m})^2}} \times \frac{1}{\sqrt{\sum_{t=1}^T (e_{\text{EMG},m}(t) - \bar{e}_{\text{EMG},m})^2}}, \quad (12)$$

where $a_{\text{sim},m}(t)$ denotes the gait-cycle-normalized simulated activation envelope of muscle m at normalized time t , $e_{\text{EMG},m}(t)$ denotes the corresponding processed human EMG envelope, and $\bar{a}_{\text{sim},m}$ and $\bar{e}_{\text{EMG},m}$ denote their temporal means. We report both per-muscle correlations and the average correlation across the selected muscles:

$$r_{\text{EMG}} = \frac{1}{M} \sum_{m=1}^M r_m, \quad (13)$$

where M is the number of muscles included in the comparison.

Higher r_m indicates stronger agreement in waveform shape after amplitude normalization and phase optimization, not agreement in absolute activation or timing. This metric should be interpreted as EMG-envelope similarity rather than direct proof of identical physiological control. Lower values may reflect alternative but mechanically valid recruitment strategies permitted by muscle redundancy. Following MuscleMimic [20], EMG similarity complements task success and kinematic tracking rather than serving as a unique ground-truth measure of internal control fidelity. The available results do not establish matched behavioral coverage between failed and successful policies; the scores alone cannot distinguish sustained task execution from an unsuccessful episode's recorded motion.

C. Study Coverage and Comparison Scope

The five paradigms serve complementary questions. Table XV identifies the comparisons reported in this paper. Motion-prior and expert-informed studies use additional information beyond task rewards, so their results should be compared within the stated study rather than treated as a fully populated full-suite leaderboard.

D. Aggregation and Ranking Interpretation

Let $s_{m,t}$ be the reported training-environment SR in percentage points for method m and task t . For a task set $\mathcal{T}$ we report

$$\bar{s}_m(\mathcal{T}) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} s_{m,t}, \quad C_m = \sum_{t=1}^{22} \mathbb{I}[s_{m,t} > 0]. \quad (14)$$

All-task aggregation weights tasks equally, so the 10 interaction tasks contribute more total weight than either six-task family. Family means are reported separately. Nonzero coverage measures breadth of observed success, not reliable mastery. Raw rewards with task-specific scales are not averaged into a cross-task score.

For perturbation type k with grid $x_0 < \dots < x_L$ and reported success $y_{m,t,k,j}$, normalized area is

$$A_{m,t,k} = \frac{\sum_{j=0}^{L-1} (x_{j+1} - x_j)(y_{m,t,k,j} + y_{m,t,k,j+1})/2}{x_L - x_0}. \quad (15)$$

Action and dynamics grids are (0,0.05,0.10,0.15,0.20); the observation grid is (0,0.02,0.05,0.08,0.10). We average A equally over the 22 tasks and then equally over the three perturbation types. This summarizes the existing robustness metric and is not an eighth metric. It depends on the specified ranges and weights and is kept separate from $\text{SR}_{\text{train-env}}$.

The ranking changes with the question. DynSyn-SAC leads mean training-environment SR and observation-sweep area; DepRL leads nonzero coverage, action/dynamics areas, and the equal-weight mean area. These are descriptive comparisons of reported point estimates, not claims of statistically significant differences.

TABLE XII

BENCHMARK PERFORMANCE AND ROBUSTNESS EVALUATION: STABILIZATION. TRAIN-ENV. SR RETAINS NATIVE TRAINING NOISE; ROBUSTNESS COLUMNS REPORT SEPARATE PERTURBATION SWEEPS (APP. B).

Task	Algorithm	Activation	Robustness (%)															Max	Train-env.	$\log_{10}$ smooth
		Cost	Action					Observation					Dynamic					Steps ($\times 10^7$)	SR (%)	(rad ² /s ⁶)
			0	0.05	0.1	0.15	0.2	0	0.02	0.05	0.08	0.10	0	0.05	0.1	0.15	0.2			
stand	DepRL	0.4488	94	94	90	96	98	100	12	0	0	0	98	48	28	12	4	3.74	86	3.152
	SAC	0.0484	100	100	100	100	96	100	0	0	0	0	100	100	100	100	100	7.13	100	1.875
	PPO	0.3691	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.97	0	5.479
	DynSyn-SAC	0.2859	52	42	56	46	36	46	46	22	30	14	40	32	40	30	30	5.44	52	2.104
powerlift	DepRL	0.3356	68	68	64	56	52	62	62	38	4	0	66	70	52	42	38	4.14	38	2.631
	SAC	0.2106	98	94	94	88	80	96	0	0	0	0	98	94	88	64	2	6.78	90	2.825
	PPO	0.3955	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.76	0	5.042
	DynSyn-SAC	0.3631	100	100	100	98	92	100	100	96	94	90	100	98	94	90	82	8.50	100	5.781
squat	DepRL	0.3781	44	46	38	32	30	24	56	34	62	26	46	38	12	20	10	14.02	42	2.984
	SAC	0.1184	0	0	0	4	0	0	0	0	0	0	4	0	0	0	0	4.79	0	1.708
	PPO	0.3455	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.01	0	5.390
	DynSyn-SAC	0.1955	48	38	32	26	22	50	44	38	36	30	48	42	32	26	18	7.00	42	6.503
sit	DepRL	0.3327	96	100	100	98	92	92	100	100	100	94	100	92	96	100	90	4.88	96	2.872
	SAC	0.1164	80	80	74	70	64	84	0	0	0	0	78	100	76	70	52	7.39	64	3.204
	PPO	0.4517	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.37	0	5.350
	DynSyn-SAC	0.3302	80	72	70	68	60	80	80	84	82	78	92	84	78	60	26	9.27	82	3.145
singlestand	DepRL	0.3977	14	16	28	18	10	24	30	8	0	0	12	28	6	8	12	2.10	14	2.854
	SAC	0.0849	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4.33	0	1.954
	PPO	0.3981	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.84	0	5.436
	DynSyn-SAC	0.3845	24	28	24	20	18	26	22	16	0	0	22	18	8	0	0	10.00	24	2.931
balance	DepRL	0.3775	60	40	26	20	14	56	0	0	0	0	46	12	36	22	18	8.86	26	2.822
	SAC	0.1459	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8.11	0	2.547
	PPO	0.3859	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.25	0	4.861
	DynSyn-SAC	0.2735	40	38	20	16	14	54	40	8	2	0	52	8	0	0	0	10.50	44	6.066

TABLE XIII

BENCHMARK PERFORMANCE AND ROBUSTNESS EVALUATION: LOCOMOTION. TRAIN-ENV. SR RETAINS NATIVE TRAINING NOISE; ROBUSTNESS COLUMNS REPORT SEPARATE PERTURBATION SWEEPS (APP. B).

Task	Algorithm	Activation Cost	Robustness(%)															Max Steps($\times 10^7$)	Train-env. SR (%)	$\log_{10}$ smooth (rad ² /s ⁶)
			Action					Observation					Dynamic							
			0	0.05	0.1	0.15	0.2	0	0.02	0.05	0.08	0.10	0	0.05	0.1	0.15	0.2			
walk forward	DepRL	0.4206	28	28	32	32	44	34	32	46	24	30	40	20	12	8	6	3.32	64	3.151
	SAC	0.1427	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9.88	0	3.262
	PPO	0.3940	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.42	0	5.558
	DynSyn-SAC	0.2771	82	88	68	60	48	82	82	80	76	76	82	76	76	64	8	5.00	76	3.417
run	DepRL	0.3354	30	22	32	36	42	18	18	30	32	26	32	16	12	8	2	6.72	18	3.343
	SAC	0.0940	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.13	0	2.854
	PPO	0.3696	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.16	0	5.524
	DynSyn-SAC	0.3221	96	94	84	70	68	96	94	90	86	80	96	88	2	2	0	10.00	96	3.435
walk turn	DepRL	0.3385	46	60	62	66	68	34	40	58	40	26	48	46	40	36	36	6.38	42	3.057
	SAC	0.1571	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.99	0	2.610
	PPO	0.3747	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.05	0	5.331
	DynSyn-SAC	0.3112	96	88	84	78	74	94	90	90	88	84	92	24	0	0	6	8.00	86	3.248
jump	DepRL	0.3616	8	24	20	12	10	18	18	8	4	0	14	10	12	6	0	6.89	10	3.188
	SAC	0.1342	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3.92	0	2.853
	PPO	0.3880	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.07	0	5.640
	DynSyn-SAC	0.2700	70	40	8	0	0	62	54	50	48	42	42	30	2	2	0	8.00	54	3.449
sidestep	DepRL	0.3888	40	38	62	56	32	34	12	2	0	0	24	22	18	10	6	4.38	28	3.349
	SAC	0.1045	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.000032	0	3.166
	PPO	0.3729	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4.92	0	5.408
	DynSyn-SAC	0.3630	74	66	54	42	30	74	78	74	78	54	64	72	60	40	40	5.58	76	3.223
crawl	DepRL	0.3540	100	100	100	100	98	96	100	100	100	100	100	100	98	100	100	5.12	86	3.158
	SAC	0.0338	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.87	0	2.516
	PPO	0.4036	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.34	0	5.674
	DynSyn-SAC	0.2766	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	10.00	100	3.444

TABLE XIV

BENCHMARK PERFORMANCE AND ROBUSTNESS EVALUATION: INTERACTION WITH ENVIRONMENT. TRAIN-ENV. SR RETAINS NATIVE TRAINING NOISE; ROBUSTNESS COLUMNS REPORT SEPARATE PERTURBATION SWEEPS (APP. B).

Task	Algorithm	Activation Cost	Robustness (%)															Max Steps ($\times 10^7$)	Train-env. SR (%)	$\log_{10}$ smooth (rad ² /s ⁶)
			Action					Observation					Dynamic							
			0	0.05	0.1	0.15	0.2	0	0.02	0.05	0.08	0.10	0	0.05	0.1	0.15	0.2			
stair	DepRL	0.3499	52	54	38	42	40	34	38	2	0	0	32	40	24	14	4	1.42	36	2.982
	SAC	0.1993	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6.98	0	2.459
	PPO	0.3854	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.57	0	5.303
	DynSyn-SAC	0.2759	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5.92	0	2.854
catch	DepRL	0.3247	80	80	76	76	84	62	48	80	72	46	78	40	42	38	56	7.23	28	2.919
	SAC	0.1384	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.37	0	2.899
	PPO	0.3964	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3.12	0	5.300
	DynSyn-SAC	0.3109	70	64	60	60	58	60	40	34	26	22	68	68	60	54	50	7.38	66	3.134
hurdle	DepRL	0.3715	48	48	56	50	40	64	44	48	28	0	62	44	40	32	20	1.26	50	3.212
	SAC	0.1852	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4.12	0	2.816
	PPO	0.3668	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.46	0	5.398
	DynSyn-SAC	0.3717	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5.50	0	7.102
slide	DepRL	0.3330	100	22	22	16	12	100	16	4	4	0	100	8	14	4	6	11.32	100	3.106
	SAC	0.0925	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4.66	0	2.752
	PPO	0.3934	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4.13	0	5.345
	DynSyn-SAC	0.2649	0	2	2	0	0	0	0	0	0	0	0	2	2	0	0	3.83	0	2.864
steppingstones	DepRL	0.3606	42	46	44	40	38	46	42	10	20	2	54	38	24	18	18	8.32	58	2.985
	SAC	0.1647	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7.33	0	2.140
	PPO	0.4020	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.27	0	5.211
	DynSyn-SAC	0.3569	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10.00	0	6.764
polewalk	DepRL	0.4080	22	38	16	40	20	42	26	6	0	0	30	26	20	16	16	5.14	12	2.876
	SAC	0.1504	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6.10	0	1.724
	PPO	0.4274	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.87	0	5.377
	DynSyn-SAC	0.2909	98	96	98	92	90	98	0	0	0	0	96	94	94	76	16	9.00	88	3.242
reach	DepRL	0.3410	84	86	84	94	82	94	82	90	62	34	92	88	82	82	74	9.34	84	2.975
	SAC	0.1705	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5.85	0	2.677
	PPO	0.3965	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2.87	0	5.152
	DynSyn-SAC	0.2556	4	14	0	4	4	2	6	6	2	4	10	8	6	4	0	2.10	0	2.840
walkandsit	DepRL	0.3648	30	26	28	14	10	28	36	32	18	10	22	28	22	14	12	2.68	16	3.365
	SAC	0.1180	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6.54	0	2.410
	PPO	0.3951	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3.03	0	5.636
	DynSyn-SAC	0.3391	48	58	66	60	52	48	54	40	36	34	42	48	40	36	18	0.45	58	7.145
chinup	DepRL	0.3998	50	59	50	62	54	46	72	64	62	56	70	62	64	60	52	5.46	32	2.862
	SAC	0.4618	64	86	76	66	58	64	50	74	72	66	70	70	68	64	60	1.37	28	2.703
	PPO	0.3620	62	74	72	66	60	64	66	82	80	72	74	66	56	60	76	1.97	74	4.710
	DynSyn-SAC	0.2797	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10.00	0	6.989
open door	DepRL	0.3304	96	100	100	100	98	100	94	74	62	42	100	96	96	78	66	4.71	98	3.030
	SAC	0.0962	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.97	0	2.836
	PPO	0.4112	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.77	0	5.327
	DynSyn-SAC	0.2784	100	98	98	96	92	100	100	100	98	98	100	100	14	4	0	9.50	100	6.461

TABLE XV

REPORTED EVALUATION COVERAGE. THE FULL-SUITE RANKING AND FOCUSED DIAGNOSTIC STUDIES ANSWER DIFFERENT QUESTIONS.

Paradigm / methods	Reported scope	Additional information	Metrics and question
Reward-based: PPO, SAC, DepRL, DynSyn-SAC	All 22 tasks	Task rewards and model observations	Training-environment SR, reward, learning steps, robustness, activation cost and jerk; cross-task comparison
Agentic reward tuning	Fig. 14, Tab. XVII; walking EMG	Reasoner updates to reward coefficients	Perturbed survival, tracking, activation, EMG; whether retuning improves control
Latent-action RL	Fig. 15, Tab. XVIII; walking EMG	Expert-trained encoder/decoder	Learning and robustness, AE MSE, PCA variance, EMG; effects of compression
Imitation prior: Muscle-Mimic	Stand, jump, walk, run, stairs in Fig. 3; EMG comparisons	Matched motions and pre-trained policy	Prior robustness, stairs tracking mismatch and recruitment shape
Residual adaptation	Walk, run, stairs	Frozen MuscleMimic and task-specific adaptation	Paired SR and EMG; tracking, height and GRF diagnostics

TABLE XVI
MEAN NORMALIZED ROBUSTNESS AREA BY PERTURBATION TYPE
(0–100 SCALE).

Method	Action	Observation	Dynamics
DynSyn-SAC	46.66	43.37	34.75
DepRL	53.06	38.10	39.84
SAC	15.36	4.26	14.50
PPO	3.10	3.40	2.92

E. Detailed Descriptions of Reward-Based RL Baselines

In Section III, we categorized several standard and exploration-enhanced reinforcement learning methods under the reward-based RL methods. The detailed descriptions of these individual baselines are provided below:

- **PPO (Proximal Policy Optimization).** This [22] serves as the quintessential on-policy actor–critic baseline for continuous-control tasks. By employing a clipped surrogate objective function, PPO ensures conservative policy updates, preventing destructively large parameter shifts that frequently occur in highly nonlinear biomechanical simulations. While it offers excellent training stability and reliable convergence, its purely reward-driven, unstructured exploration often proves sample-inefficient when navigating the high-dimensional, highly redundant muscle excitation spaces inherent to full-body musculoskeletal humanoids.
- **SAC (Soft Actor-Critic).** This [23] is utilized as the primary off-policy baseline, distinguished by its maximum-entropy reinforcement learning framework. By jointly maximizing expected return and policy entropy, SAC explicitly encourages broad and robust exploration of the action space. This stochastic exploration strategy is particularly advantageous for musculoskeletal control: the entropy regularization prevents premature convergence to sub-optimal local minima—a common pathology in muscle-driven systems where multiple distinct muscle activation patterns can yield identical kinematic joint torques. Furthermore, its off-policy nature and replay buffer significantly improve sample efficiency over on-policy methods.
- **DepRL.** This [24] represents a specialized, exploration-enhanced reinforcement learning baseline tailored for musculoskeletal systems. Traditional independent Gaussian action noise often fails to activate muscles in a biologically coordinated manner, typically resulting in conflicting co-contractions and “stiff” joint behaviors. DepRL addresses this by augmenting the RL policy with Differential Extrinsic Plasticity (DEP)—an unsupervised, self-organizing exploration mechanism. Instead of injecting blind stochastic noise, the DEP controller dynamically updates a sensorimotor mapping matrix based on the temporal correlations between recent motor commands and the derivatives of proprioceptive feedback. This generates highly structured, mechanically

grounded exploratory actions that intrinsically leverage the agent’s physical embodiment. By superimposing these synergistic exploration patterns onto the RL policy outputs, DepRL effectively mitigates stiff co-contractions and significantly accelerates the discovery of functional locomotion gaits in highly overactuated models.

- **DynSyn-SAC.** This [4] is a state-of-the-art action-representation algorithm designed specifically for efficient learning and control in overactuated embodied systems. Built upon the maximum-entropy SAC framework, DynSyn-SAC introduces a dynamical synergistic structure directly into the network architecture. Specifically, it groups functionally related actuators and processes state observations into a lower-dimensional latent representation. A dedicated *DynSyn Layer* then maps this latent output to full-body muscle activations using group-shared synergistic weights. To ensure stable exploration during the early phases of learning, the algorithm employs a linear scheduling mechanism that dynamically bounds and scales the amplitude of these synergistic weights over the course of training. This hierarchical approach allows the policy to effectively predict group-level synergistic actions alongside state-dependent, muscle-level fine-tuning, thereby significantly accelerating sample efficiency in high-dimensional musculoskeletal models.

F. Detailed Reward Curves for Reward-Based RL Baselines

In this appendix, we present the detailed training results of the reward-based RL baselines. The reward curves for all 22 tasks are shown in Figure 11.

G. Detailed Muscle Activation Analysis: 416 vs. 700 Muscle Models

The anatomy-resolution study examines how available muscle groups affect recruitment during the **powerlift** task. This comparison is separate from the shared 416-muscle benchmark protocol and does not establish that increased anatomical detail necessarily improves physiological accuracy. To provide deeper insight into these internal control differences, we present the time-step resolved muscle activation heatmaps for both the 416-muscle (Fig. 12) and the highly detailed 700-muscle (Fig. 13) models.

Biomechanical Analysis. The heatmaps show different activation distributions in the two models. In the 416-muscle model, several arm and shoulder groups exhibit sustained high activation. The 700-muscle model additionally recruits neck, deep-trunk, and scapular groups and shows lower activation in some shared upper-limb muscles. This pattern is consistent with alternative load-support pathways becoming available. Activation heatmaps alone do not measure mechanical load transfer, clinical correctness, or human physiological fidelity; these interpretations require independent force and human-reference validation.

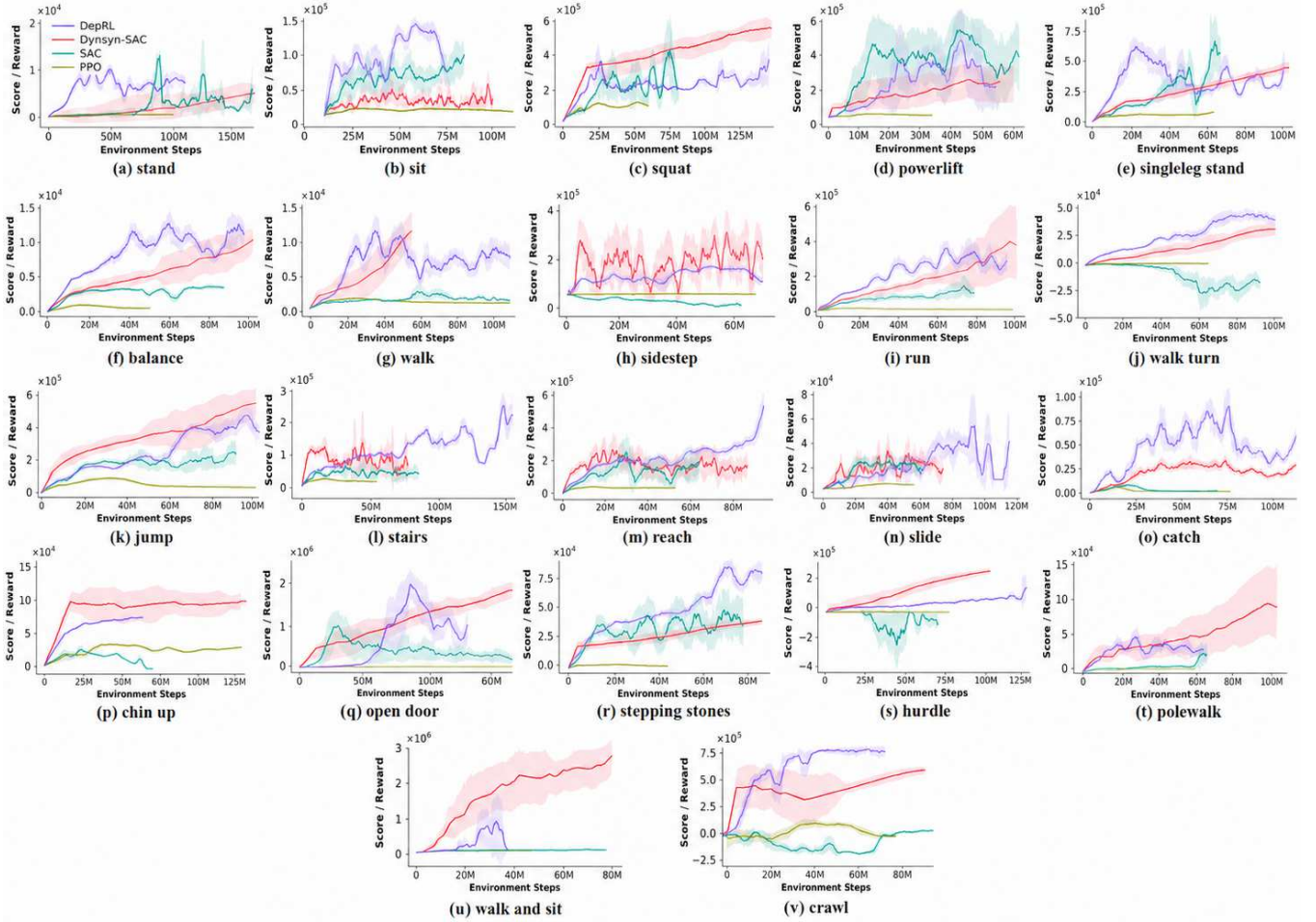


Fig. 11. **Learning curves for all 22 tasks.** This figure illustrates the detailed reward trajectories of the pure reinforcement learning baselines across the complete suite of 22 tasks during training.

H. Agentic Reward Tuning

In our framework, we introduce *Agentic Reward Tuning*, a training mechanism that autonomously adjusts reward weights during reinforcement learning according to the agent’s observed performance. Instead of relying on a fully fixed reward design, the reward composition is periodically refined by an external reasoning module based on recent gait quality, balance stability, and task-progress statistics.

Formally, let $\mathbf{r}_t$ denote the vector of sub-rewards at timestep t , and let $\mathbf{w}_t$ denote the corresponding reward weights. The dense reward is computed as:

$$R_t = \sum_i w_{i,t} r_{i,t}. \quad (16)$$

During training, the reward weights are not treated as static hyperparameters. Every K environment steps, the system aggregates recent performance statistics:

$$\mathbf{s}_{\text{stats}} = \text{Aggregate}(\{\mathbf{o}_\tau, \mathbf{r}_\tau\}_{\tau=t-K}^t), \quad (17)$$

where $\mathbf{s}_{\text{stats}}$ contains task-relevant indicators such as forward velocity, torso pitch, posture stability, and tracking quality.

The reasoning module then updates the reward weights according to the current reward configuration and the collected statistics:

$$\mathbf{w}_{t+1} = \text{Reasoner}(\mathbf{w}_t, \mathbf{s}_{\text{stats}}), \quad (18)$$

with the update magnitude constrained to avoid destabilizing policy optimization:

$$\|\mathbf{w}_{t+1} - \mathbf{w}_t\|_\infty \leq \epsilon. \quad (19)$$

For example, if the agent exhibits insufficient forward speed, the tuning module increases the velocity-tracking weight. If the agent develops a hunched posture, the weights of uprightness and anti-hunching rewards are increased. In this way, Agentic Reward Tuning acts as an adaptive reward-shaping mechanism that continuously emphasizes the currently deficient aspects of the learned behavior.

In implementation, the tuning process is triggered periodically during environment rollout. Recent statistics, such as average forward speed and torso pitch, are cached over a fixed interval. A reasoning module analyzes these statistics and returns an updated reward-weight dictionary. To support parallel training, the updated weights are written to a

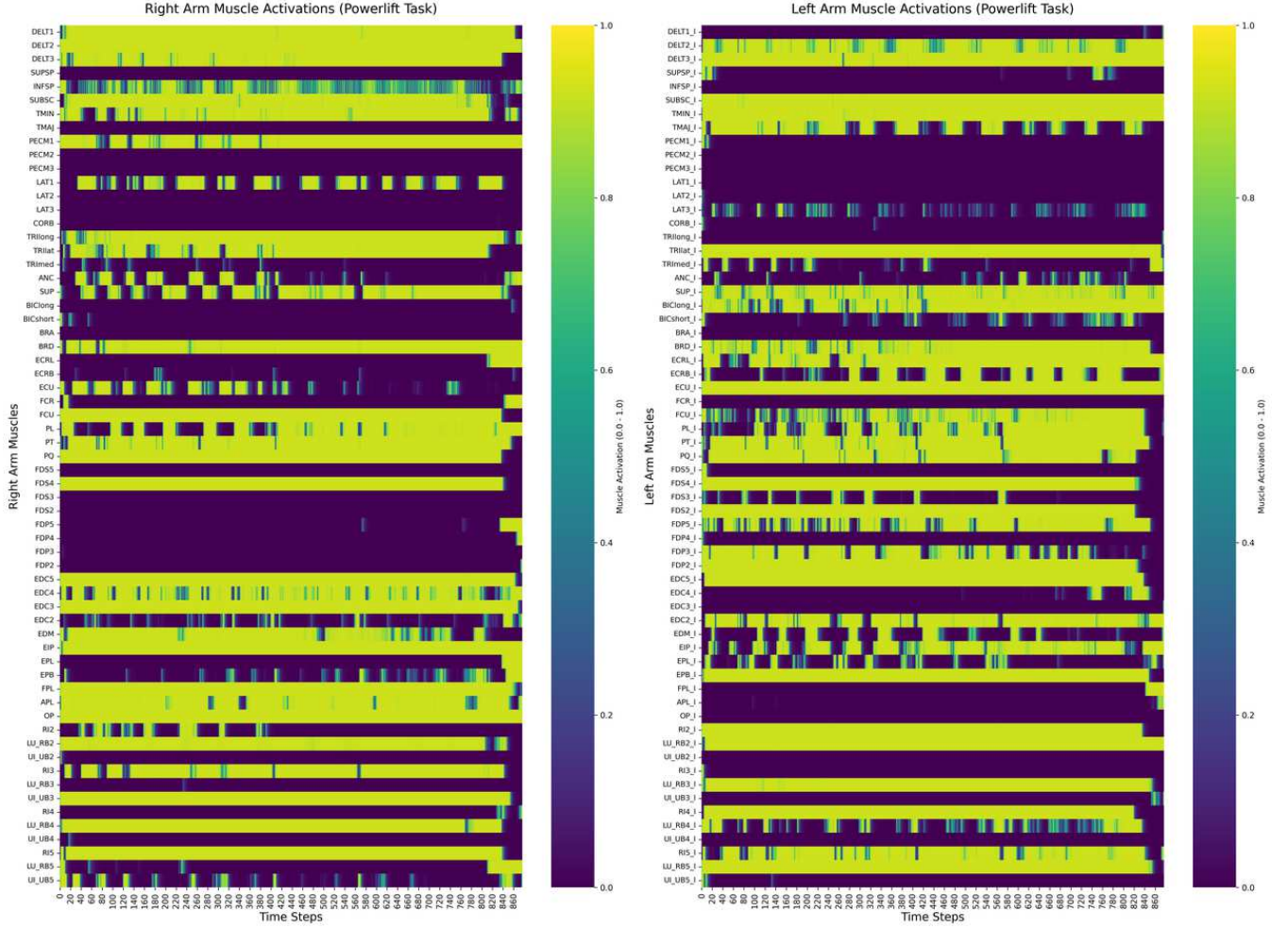


Fig. 12. **Upper limb muscle activation heatmap for the 416-muscle model (Powerlift).** The heatmap displays the intense, sustained activations of the available arm and shoulder musculature required to stabilize the barbell overhead. Note that this model lacks the detailed deep axial stabilizers of the neck and spine.

shared file and synchronized across environments. A file-lock mechanism is used to prevent multiple workers from simultaneously calling the reasoning module, and fallback exception handling ensures that training continues even when the external call fails.

The reward basis remains fixed throughout tuning, so this experiment evaluates coefficient adaptation rather than reward-term generation. A bound on individual updates does not by itself constrain cumulative weight scale. In addition to the original online trajectories in Fig. 14, we therefore evaluate a scale-controlled variant that constrains the adjustable-weight norm to 40, removing proportional reward-scale inflation as a confound.

Table XVII reveals a trade-off: improved tracking and reduced activation coexist with shorter survival and a larger aggregate gap. Thus, the negative result is not explained solely by reward-scale inflation. It concerns this tested coefficient-adaptation procedure and should not be generalized to all automated reward optimization methods.

TABLE XVII

SCALE-CONTROLLED REWARD ADAPTATION. THE ADJUSTABLE-WEIGHT NORM IS CONSTRAINED TO 40. THESE DIAGNOSTICS COMPARE FIXED COEFFICIENTS WITH ADAPTED COEFFICIENTS; THE AGGREGATE GAP IS AN AUXILIARY DIAGNOSTIC, NOT AN EIGHTH BENCHMARK METRIC.

Diagnostic	Fixed	Adapted
Tracking RMSE ↓	0.4446	0.4355
Activation ↓	0.4769	0.3999
Survival steps ↑	417	246.6
Aggregate benchmark gap ↓	13.42	14.58

I. Latent-Action RL

The *Latent-Action RL* framework integrates expert-informed action filtering with reinforcement learning to guide musculoskeletal humanoid control. In the hard mode used in this work, the policy is not allowed to directly execute arbitrary high-dimensional muscle actions. Instead, each policy action is first projected through a pretrained anatomical

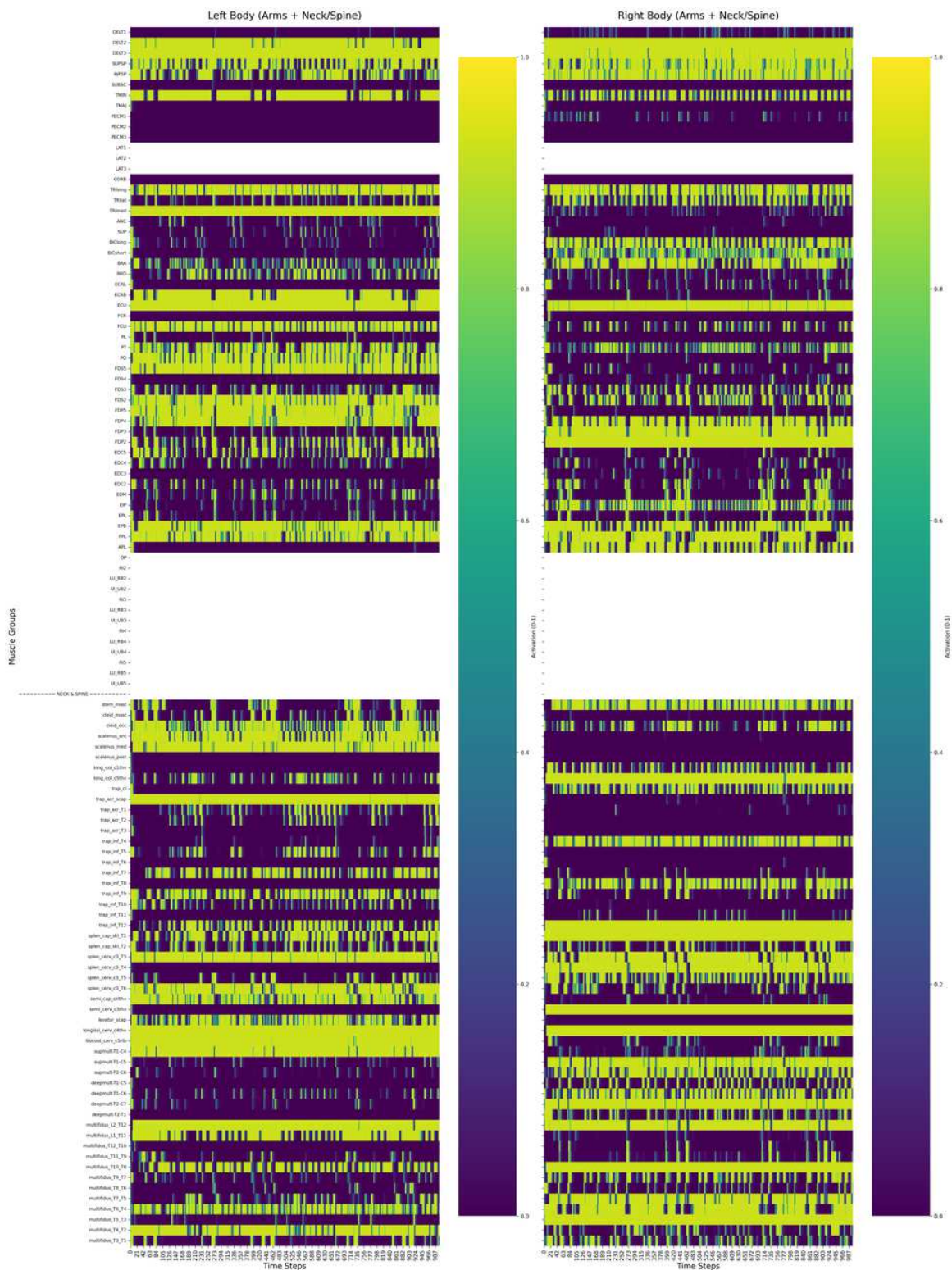


Fig. 13. **Expanded muscle activation heatmap for the 700-muscle model (Powerlift).** The top section includes the upper limb muscles, where the distinct "blank gaps" represent the fine-grained scapular and arm muscles exclusive to the 700-model. The bottom section explicitly details the heavy recruitment of the newly available neck and spine axial stabilizers (e.g., *multifidus*, *splenius*).

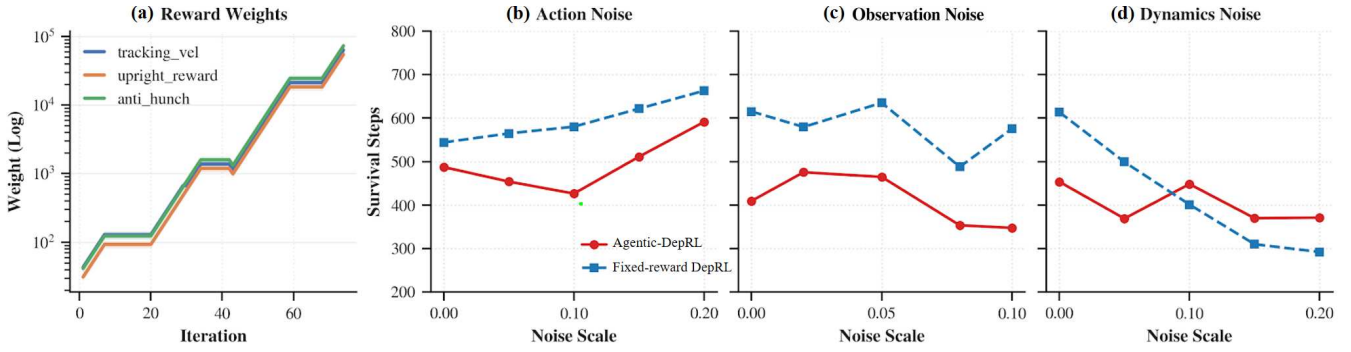


Fig. 14. **Online LLM-guided reward-weight tuning does not consistently improve robustness over fixed-reward DepRL.** (a) Evolution of reward weights updated online during training; the reward terms are fixed and only scalar weights are modified. (b–d) Survival steps under **action**, **observation**, and **dynamics** perturbations for Agentic-DepRL and fixed-reward DepRL. Both variants use the same DepRL [24] base algorithm, training budget, and policy architecture. Survival is capped at 1000 steps; each point averages 50 held-out evaluation seeds. More details are given in App. B and App. H.

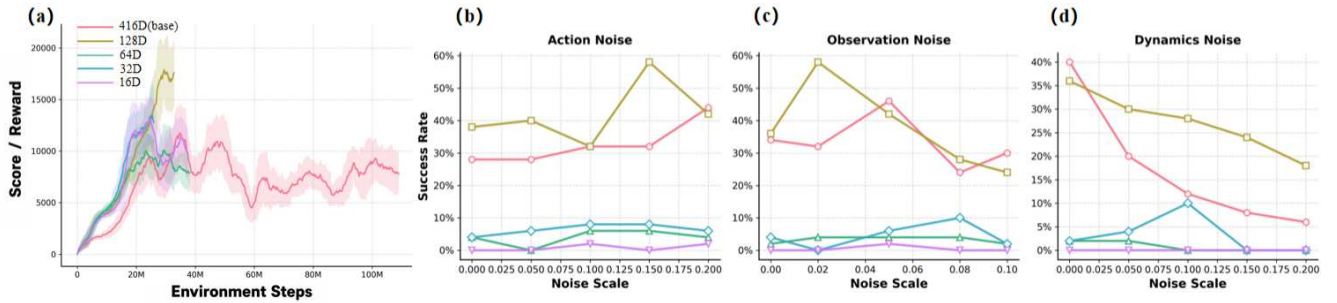


Fig. 15. **Moderate action-space compression helps, but aggressive compression loses expressivity.** Reducing the 416-dimensional muscle-control space to 128 dimensions improves reward learning and several robustness settings, whereas stronger compression to 64, 32, or 16 dimensions substantially degrades performance (details in App. I).

encoder-decoder module, and the final executable action is constrained to remain within a bounded neighborhood of the expert-informed action. This design reduces unstable exploration in the high-dimensional muscle action space while preserving the adaptability of reinforcement learning.

1. Expert Teacher Pretraining: An expert teacher model is first trained using behavior cloning on high-quality demonstration data. The model adopts an encoder-decoder architecture:

- **Encoder:** maps the expert muscle action a_t to a low-dimensional latent intent z_t , which captures the underlying control intention.
- **Decoder (Full Anatomical Transformer):** reconstructs the muscle action conditioned on z_t and the current musculoskeletal state, including muscle priors, muscle states, and joint moment information.

Formally, for a batch of expert actions A and associated anatomical features (P, S, M) , the reconstructed action is given by:

$$\hat{A} = \text{Decoder}(\text{Encoder}(A), P, S, M), \quad (20)$$

where P denotes muscle-level physical priors, S denotes muscle state features such as muscle length, velocity, force, and activation, and M denotes the muscle-joint moment

matrix. The behavior cloning objective is:

$$\mathcal{L}_{\text{BC}} = \frac{1}{N} \sum_i \left\| \hat{A}_i - \frac{A_i + 1}{2} \right\|^2. \quad (21)$$

Here, the transformation $(A_i + 1)/2$ maps the original normalized action range $[-1, 1]$ to the muscle activation range $[0, 1]$.

2. Hard-mode Action Filtering in RL: During reinforcement learning, the policy outputs a raw action $a_t^{\text{RL}} \in [-1, 1]^M$. This action is first normalized to the muscle activation range:

$$a_{t,01}^{\text{RL}} = \frac{a_t^{\text{RL}} + 1}{2}. \quad (22)$$

The pretrained encoder and decoder then generate an expert-informed filtered action:

$$\hat{a}_t = \text{Decoder}(\text{Encoder}(a_{t,01}^{\text{RL}}), P_t, S_t, M_t), \quad (23)$$

where $\hat{a}_t \in [0, 1]^M$ represents the anatomical action prior predicted by the transformer decoder.

In hard mode, the final executable action is clipped within a fixed-radius action tube around the expert-informed action:

$$a_{t,01}^{\text{env}} = \text{clip}(a_{t,01}^{\text{RL}}, \hat{a}_t - \delta, \hat{a}_t + \delta), \quad (24)$$

where δ is the tube radius controlling the maximum allowed deviation from the anatomical prior. The action is then

mapped back to the environment action range:

$$a_t^{\text{env}} = 2a_{t,01}^{\text{env}} - 1. \quad (25)$$

This constraint limits deviation from the expert-informed action manifold while permitting local exploration; physiological plausibility still requires evaluation.

3. Deviation Penalty: To further discourage unnecessary deviation from the expert-informed action, the hard mode introduces a penalty based on the violation between the original RL action and the clipped safe action:

$$\mathcal{L}_{\text{dev}} = \|a_{t,01}^{\text{RL}} - a_{t,01}^{\text{env}}\|^2. \quad (26)$$

The environment reward is modified as:

$$R'_t = R_t - \lambda_t \mathcal{L}_{\text{dev}}, \quad (27)$$

where λ_t is a decaying coefficient. At the early stage of training, the deviation penalty strongly guides the policy toward the expert action manifold. As training progresses, the penalty gradually decreases, allowing the policy to adapt more freely to task-specific objectives.

4. Transformer-based Muscle Attention: The Full Anatomical Transformer models inter-muscle dependencies through stacked Muscle Attention Blocks. Each block consists of multi-head self-attention, layer normalization, feed-forward transformations, and residual connections. By attending across muscles, the decoder captures global muscle synergy patterns and joint-level coordination, which are difficult to model using independent muscle-wise mappings.

5. Overall Principle: In hard mode, Latent-Action RL can be viewed as reinforcement learning inside an expert-guided anatomical action tube. The pretrained encoder-decoder provides a biologically plausible action manifold, while reinforcement learning performs bounded exploration around this manifold. This mechanism provides an action-structure prior whose utility depends on retained expressivity, as tested below.

Representation diagnostics. We additionally evaluate held-out autoencoder reconstruction MSE and PCA explained variance as action dimension is reduced. Across the tested dimensions from 128 to 16, MSE increases monotonically from 0.0084 to 0.0188, while explained variance decreases from 89.4% to 65.4%. These measures probe representation quality independently of RL return. Their joint degradation supports an expressivity bottleneck at aggressive compression levels; it does not prove that dimensionality alone explains every policy failure.

TABLE XVIII
ENDPOINTS OF THE ACTION-COMPRESSION DIAGNOSTICS.

Diagnostic	128 dimensions	16 dimensions
Held-out AE MSE ↓	0.0084	0.0188
PCA variance (%) ↑	89.4	65.4

J. EMG Data Processing and Comparative Analysis

This section describes the processing and comparative analysis of electromyography (EMG) signals for evaluating simulated and RL-generated muscle activations against human reference data.

1. Data Normalization and Gait Cycle Alignment EMG signals are first normalized to the gait cycle. Each signal is resampled to 101 points per cycle using cubic interpolation:

$$\tilde{x} = \text{interp1d}(x_{\text{old}}, x_{\text{raw}}, \text{kind}='cubic')(x_{\text{new}}), \quad (28)$$

where x_{old} represents the original sample points and x_{new} is the normalized gait percentage from 0% to 100%. Signals are then min-max normalized to $[0, 1]$:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min} + \varepsilon}, \quad (29)$$

with ε a small constant to prevent division by zero.

2. Multi-Cycle Averaging For EMG recordings with multiple gait cycles, individual cycles are extracted by detecting peaks corresponding to gait events (e.g., heel strikes). Each cycle is normalized and resampled, then averaged across cycles to produce a mean activation profile. The standard deviation across cycles is also computed and normalized by the peak-to-peak amplitude to quantify variability:

$$\sigma_{\text{norm}} = \frac{\text{std}(x_{\text{cycles}})}{x_{\max} - x_{\min}}. \quad (30)$$

3. Simulation-Human Alignment Simulated EMG signals are aligned with the human reference by cyclically shifting the normalized simulation profile to maximize the Pearson correlation coefficient r :

$$r = \frac{\text{COV}(\tilde{x}_{\text{sim}}, \tilde{x}_{\text{human}})}{\sigma_{\text{sim}} \sigma_{\text{human}}}. \quad (31)$$

The shift that maximizes r is applied to both the mean and standard deviation of the simulated signal, providing a phase-optimized waveform comparison. Because this alignment maximizes correlation, the resulting score does not independently validate absolute phase timing relative to heel strike.

4. Comparative Analysis For each target muscle, the correlation coefficient r quantifies waveform-shape agreement after phase optimization between simulated activation and human EMG. The normalized standard deviation summarizes variability across normalized cycles. Per-muscle normalization removes absolute magnitude information, and phase optimization limits interpretation of absolute timing. Correlation therefore measures relative temporal recruitment shape rather than raw EMG magnitude, muscle force, or unique physiological correctness. Subject-specific recruitment and muscle redundancy permit alternative mechanically valid solutions. The extended running/stairs results in Tab. IV are descriptive averages across 12 matched muscles, with task-dependent method rankings.

K. Muscle Recruitment Categorization

To analyze whole-body muscle recruitment patterns, we categorize the 416 muscle actuators into anatomically and functionally meaningful muscle families. This grouping converts the high-dimensional muscle activation vector into an interpretable recruitment profile, allowing us to quantify how different body regions contribute to each learned behavior.

For each trained policy, we perform a rollout in the corresponding musculoskeletal environment and record the activation of each muscle actuator at every simulation timestep. Let $a_{m,t}$ denote the activation of the m -th muscle at timestep t , where $m \in \{1, \dots, M\}$ and $M = 416$. The mean activation of each muscle over one rollout is computed as:

$$\bar{a}_m = \frac{1}{T} \sum_{t=1}^T a_{m,t}, \quad (32)$$

where T denotes the number of simulated timesteps before termination or the predefined maximum rollout length.

Each muscle actuator is assigned to an anatomical family according to its actuator name. Specifically, each category contains a set of predefined name keywords. A muscle is assigned to category C_k if its actuator name contains at least one keyword belonging to that category:

$$m \in C_k \iff \exists q \in \mathcal{Q}_k \text{ s.t. } q \subseteq \text{name}(m), \quad (33)$$

where $\mathcal{Q}_k$ denotes the keyword set of the k -th muscle category.

The absolute activation mass of each muscle family is then computed by summing the mean activations of all muscles assigned to that family:

$$A_k = \sum_{m \in C_k} \bar{a}_m. \quad (34)$$

The total activation mass across all muscle families is:

$$A_{\text{total}} = \sum_{k=1}^K A_k. \quad (35)$$

Finally, the relative contribution of each muscle family is reported as:

$$P_k = \frac{A_k}{A_{\text{total}}} \times 100\%. \quad (36)$$

The full anatomical grouping rule used in our analysis is summarized in Table XIX. This categorization covers the major trunk, upper-limb, lower-limb, and hand muscle groups in the 416-muscle model.

Muscles that do not match any predefined keyword are assigned to an *Unclassified* group. During analysis, we monitor the activation mass of this group to verify that the predefined anatomical categories cover the majority of meaningful muscle activity. If the activation mass of the unclassified group is negligible, the resulting recruitment distribution can be considered a reliable summary of whole-body muscle usage.

This categorization allows us to compare policies not only by task reward or success rate, but also by biomechanical recruitment strategy. For example, an efficient powerlifting

policy should exhibit stronger recruitment of the posterior chain and trunk stabilizers, while a locomotion policy should show coordinated contributions from hip extensors, quadriceps, hamstrings, and plantar flexors.

L. Quantitative Load Intervention

Using the family activation mass A_k defined above, we aggregate axial/core and upper-extremity activation and compare their ratio and their shares of total activation. Under comparable task success, the axial/core-to-upper-extremity ratio increases monotonically across increasing barbell loads from 1.42 $\rightarrow$ 1.52 $\rightarrow$ 2.19 $\rightarrow$ 2.27, with endpoints at 0.05 and 10 kg. The endpoint change is approximately 60%. Table XX reports the loads for which the regional shares are available.

Uniformly scaling every muscle's activation would preserve these shares and ratios. Their change therefore indicates redistribution of recruitment under increasing load. In particular, the axial/core share peaks at 5 kg among the reported values and decreases at 10 kg, even though the regional ratio continues to rise. These activation-based summaries support demand-dependent coordination, but do not identify individual muscle forces or establish a unique human recruitment solution.

M. Detailed Descriptions of Residual RL Policy

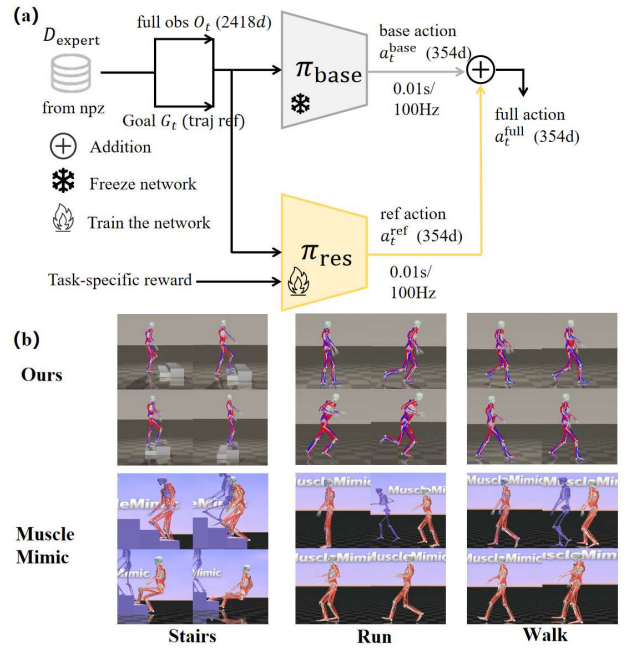


Fig. 16. **Residual RL Architecture and Performance.** (a) Overview of the residual framework, where the final action is the sum of a frozen imitation prior policy [20] and a task-optimized residual network. (b) Visual comparison of **stair-climbing**, **running** and **walking** gaits. **Blue** skeletons denote the reference imitation prior; colored trajectories illustrate the residual policy adaptations.

To adapt the base tracking policy for robust locomotion and overcome the limitations of strict kinematic imitation across various terrains, we evaluate a task-specific adaptation pipeline built on a frozen MuscleMimic policy and

TABLE XIX

ANATOMICAL MUSCLE-FAMILY CATEGORIES USED FOR WHOLE-BODY RECRUITMENT ANALYSIS. EACH ACTUATOR IS ASSIGNED TO A CATEGORY ACCORDING TO WHETHER ITS NAME CONTAINS ONE OF THE CORRESPONDING KEYWORDS.

Body Region	Muscle Category	Actuator Name Keywords
Core and Back	Paraspinals (Deep)	Ps_
	Iliocostalis	IL_
	Longissimus (Erector Spinae)	LTPt, LTpL
	Abdominals	rect_abd, EO, IO
	Other Trunk/Back	LAT, LD, QL, MF_m, TRAP, RHOM
Upper Extremity	Pectorals	PEC
	Deltoids	DELT
	Rotator Cuff	SUPSP, INFSP, SUBSC, TMIN, TMAJ, CORB
	Elbow Flexors	BIC, BRA, BRD
	Elbow Extensors	TRI, ANC
	Forearm Pronators/Supinators	SUP, PT, PQ
	Wrist/Finger Flexors	FCR, FCU, PL, FDS, FDP, FPL
	Wrist/Finger Extensors	ECR, ECU, EDC, EDM, EIP, EPL, EPB, APL
Lower Extremity	Hand Intrinsic	OP, RI2, RI3, RI4, RI5, LU_, UI_
	Glutes (Hip Extensors)	GMAX, GMED, GMIN, PIR, GEM, OINT, QFM, glmax, glmed, glmin, piri
	Quadriceps (Knee Extensors)	VAS, RF, VI, VL, VM, vasint, vaslat, vasmed, recfem
	Hip Flexors/Adductors	ILIACUS, PSOAS, ADD, PECT, GRAC, TFL, iliacus, psoas, add, grac, tfl, sart
	Hamstrings (Knee Flexors)	BFLH, BFSL, SEMIM, SEMIT, bflh, bfsh, semiten, semimem
	Calves (Plantar Flexors)	GAS, SOL, soleus, gaslat, gasmed
	Other Leg/Foot	TA, EDL, EHL, FDL, FHL, TP, PER, edl, ehl, fdl, fhl, perbrev, tibant, tibpost, perlong

TABLE XX

POWERLIFT ACTIVATION REDISTRIBUTION. SHARES ARE PERCENTAGES OF TOTAL ACTIVATION MASS; THE RATIO IS DIMENSIONLESS.

Barbell load (kg)	0.05	5	10
Axial/core share (%)	47.3	57.1	54.5
Upper-extremity share (%)	33.3	26.1	24.0
Axial/core to upper ratio	1.42	2.19	2.27

TABLE XXI

COMPARISON ON THE WALK TASK.

Method	SR(%)	Pearson correlation $r \uparrow$		
		SoleusMed.	Bic.Fem.	Semitend.
MuscleMimic [20]	100	0.71	0.69	0.63
Residual	100	0.75	0.80	0.67

TABLE XXII

COMPARISON ON THE STAIRS CLIMBING TASK.

Method	SR (%) $\uparrow$	Max Z-Height (m) $\uparrow$	Tracking Error (rad) $\downarrow$
MuscleMimic [20]	0.0	0.94	0.1968
Residual	100	1.21	0.1079

TABLE XXIII

COMPARISON ON THE HIGH-IMPACT RUN TASK.

Method	SR (%)	Peak GRF ($\times$ Body Wt.) $\downarrow$	Tracking Error (rad) $\downarrow$
MuscleMimic [20]	90	6.40	0.2267
Residual	100	6.07	0.1338

a trainable residual network. These adaptations allow the highly overactuated musculoskeletal humanoid to overcome geometric discrepancies between the reference trajectory and the physical environment.

1) *Action Space and Residual Scaling*: Instead of allowing the residual policy to fully overwrite the base policy, we constrain the residual action space to ensure stability. The final blended action a_t sent to the musculoskeletal simulation is defined as:

$$a_t = \text{clip}(a_{\text{base}} + \alpha \cdot \text{clip}(a_{\text{res}}, -1.0, 1.0), -1.0, 1.0) \quad (37)$$

where a_t is the final blended action at time step t , a_{base} is the action inferred by the frozen base policy, a_{res} is the output of the residual network, and α is the residual scaling factor. This control ratio grants the residual policy sufficient authority to adjust muscle activations while preserving the fundamental synergistic patterns learned by the base policy. Furthermore, we deliberately disable standard kinematic penalties (e.g., action rate and activation energy penalties) to

encourage aggressive exploration, retaining only a minimal L2 regularization on the residual action ($c_{\text{res}} = -0.02$).

2) *Adaptations for Stair Navigation:* A fundamental challenge in mimicking flat-ground or mismatched reference trajectories on physical stairs is the accumulation of vertical errors. If penalized for absolute height discrepancies, the policy prematurely terminates.

Z-Axis Agnostic Imitation Objective: We address this by masking the Z-axis displacement in the root position error calculation:

$$d_{\text{root}} = \|\mathbf{p}_{xy} - \hat{\mathbf{p}}_{xy}\|_2^2 \quad (38)$$

where d_{root} is the calculated root position error, and $\mathbf{p}_{xy}$ and $\hat{\mathbf{p}}_{xy}$ are the 2D planar coordinates of the simulated and reference root positions, respectively. This height-invariant formulation allows the humanoid to step onto elevated platforms without suffering catastrophic drops in the tracking reward.

Elastic Trajectory Pacing: To prevent the policy from being dragged forward when encountering physical obstacles (e.g., tripping on a stair edge), we decouple the reference step from the physical simulation time. The environment implements an *elastic tracking* mechanism: the reference frame index advances only if the humanoid’s root deviation remains within a safe threshold ($d < 0.4$ m). If the agent lags, the trajectory progression slows down by 50%, explicitly waiting for the agent to recover its balance.

Task-Driven Progress Reward: To break the deadlock of purely conservative imitation, we augment the imitation reward with a task-driven progress reward:

$$r_{\text{prog}} = w_{\text{prog}} \left(\text{clip}(v_{\text{proj}}, 0, 2.0) + 0.5 \cdot \text{clip}(v_z, 0, 1.5) \right) \quad (39)$$

where r_{prog} is the calculated progress reward, $w_{\text{prog}} = 0.3$ is the weight coefficient for this reward, v_{proj} is the current root velocity projected onto the directional vector pointing to the future reference pelvis position (1.0s ahead), and v_z is the absolute vertical climbing velocity.

3) *Enhancements for Energy-Efficient Walking:* While navigating stairs requires aggressive spatial exploration, flat-ground walking prioritizes energy efficiency and stable transitions between stance and swing phases. Strict kinematic imitation often forces the musculoskeletal model into stiff, metabolically inefficient gaits.

Phase-Dependent Residual Authority: We dynamically modulate the residual network’s authority based on the gait phase. The swing leg requires more flexibility to adjust foot clearance, whereas the stance leg demands strict synergistic stability. The control ratio is formulated as:

$$\alpha_i(t) = \alpha_{\text{stance}} + (\alpha_{\text{swing}} - \alpha_{\text{stance}}) \cdot \mathcal{I}_{\text{swing}}(i, t) \quad (40)$$

where $\alpha_i(t)$ represents the dynamic residual scaling factor for limb i at time step t , i denotes the respective limb, $\alpha_{\text{stance}} = 0.2$ and $\alpha_{\text{swing}} = 0.6$ are the base control ratios for the stance and swing phases respectively (the latter is used for the stair

task as well), and $\mathcal{I}_{\text{swing}}(i, t)$ is a boolean indicator function derived from foot-ground contact sensors.

Metabolic-Aware Muscle Regularization: To eliminate high-frequency, non-functional muscle twitches, we reinstate a smoothed metabolic penalty specifically for the walk task:

$$p_{\text{met}} = -w_{\text{met}} \sum_{j=1}^N (a_{j,t}^2 + \lambda(a_{j,t} - a_{j,t-1})^2) \quad (41)$$

where p_{met} is the metabolic penalty, $N = 416$ is the total number of muscles, $a_{j,t}$ is the activation of muscle j at the current time step t , $a_{j,t-1}$ is the activation of the same muscle at the previous time step, $w_{\text{met}} = 0.05$ is the scaling weight for the penalty, and $\lambda = 0.1$ is the smoothing coefficient. This regularizer encourages lower activation cost and smoother activation changes; kinematic accuracy and EMG similarity are assessed separately.

4) *Adaptations for High-Impact Locomotion (Run Task):* Running fundamentally differs from walking due to the presence of an aerial flight phase. Discrepancies in ground contact timing between the reference kinematic data and the physics simulation often cause failures during impact.

Flight-Phase Propulsive Reward: During the aerial phase, we temporarily relax the tracking penalty by 50% and introduce a propulsive velocity reward:

$$r_{\text{propel}} = w_{\text{vel}} \cdot \max(0, v_x - \hat{v}_x) \cdot \mathcal{I}_{\text{air}} \quad (42)$$

where r_{propel} is the propulsive velocity reward, w_{vel} is the weight coefficient for this reward, v_x is the current forward velocity, $\hat{v}_x$ is the target running speed, and $\mathcal{I}_{\text{air}}$ indicates the flight phase. This encourages explosive muscle forces during toe-off.

Ground Reaction Force (GRF) Smoothing: High-speed impacts cause severe numerical instability in the simulation. We introduce a soft penalty for excessive vertical Ground Reaction Forces (GRF):

$$p_{\text{impact}} = -w_{\text{grf}} \cdot \max(0, F_z - F_{\text{limit}}) \quad (43)$$

where p_{impact} is the soft impact penalty, w_{grf} is the weight coefficient for the GRF penalty, F_z is the total measured vertical contact force, and F_{limit} is set to $2.5 \times$ the humanoid’s body weight. This penalty encourages moderation of landing forces; the present results do not isolate its causal contribution from the other adaptation components.

N. Extended Discussion on Limitations and Future Directions

While the Musculoskeletal Humanoid Benchmark provides a comprehensive and diverse foundation for muscle-driven control, it is essential to acknowledge several limitations in the current scope of this work. These limitations also highlight promising avenues for future research in biomechanical reinforcement learning and humanoid robotics.

1) *Absence of Dexterous Manipulation:* The current iteration of our benchmark primarily concentrates on gross motor skills, such as full-body locomotion, postural stabilization, and basic environmental interactions (e.g., pushing a door or reaching). However, one of the most defining characteristics of the human musculoskeletal system is its extraordinary capacity for fine motor control, particularly in the hands and fingers. Modeling the biomechanics of the human hand is highly complex due to the dense routing of tendons, high degrees of freedom, and intricate muscle synergies. Consequently, our benchmark currently lacks fine-grained dexterous manipulation tasks, such as in-hand object manipulation, tool use, or precise grasping. Future work should integrate high-fidelity musculoskeletal hand models into the full-body framework, enabling the study of how gross arm movements dynamically couple with fine finger manipulations.

2) *Transitioning to Compositional and Long-Horizon Tasks:* Currently, our benchmark evaluates policies on isolated, single-objective tasks (e.g., exclusively walking, or solely executing a powerlift). Real-world humanoid operation, however, inherently demands the execution of compositional and long-horizon behaviors. For instance, a realistic scenario might require the robot to walk toward a chair, pick up a barbell along the way, and subsequently perform a seated lift. Addressing such long-horizon tasks will require policies to not only master atomic skills but also smoothly transition between them without losing structural balance or experiencing catastrophic energy depletion. Future iterations of this benchmark will aim to include compositional task suites, serving as a testbed for hierarchical reinforcement learning (HRL), skill discovery, and long-term fatigue modeling in muscle-driven systems.

3) *Constraints in Comprehensive Physiological Validation:* A core motivation of musculoskeletal modeling is to generate biologically plausible behaviors. However, our physiological diagnostics compare policy-generated activation envelopes with human Electromyography (EMG) references for walking, running, and stairs, including an extended analysis across 12 matched lower-limb muscles. These comparisons leave most tasks and upper-body muscle groups unvalidated. Normalization removes absolute magnitude information, and correlation-maximizing phase alignment limits claims about absolute activation timing. Muscle redundancy and inter-subject variability further prevent treating one human recording as a unique ground-truth target. This limitation stems from a scarcity of comprehensive, open-source EMG datasets for highly dynamic or unconventional tasks (e.g., hurdle clearance, whole-body crawling, or wobble board balancing). To achieve a more universal physiological benchmark, future research must involve establishing advanced data collection pipelines—synchronizing high-density EMG with full-body motion capture—to build a diverse dataset covering a broader spectrum of human motor skills.

4) *Sim-to-Real Gap and Hardware Validation:* Finally, all evaluations within this benchmark are conducted in a

simulated physics environment. Although the simulation enforces rigorous biomechanical and physical constraints, it inevitably abstracts away certain real-world complexities, such as sensor noise, communication latency, and unmodeled actuator dynamics. Validating these muscle-driven policies on physical hardware remains a formidable challenge. Real-world musculoskeletal humanoids (e.g., tendon-driven robots) are notoriously difficult to build and maintain. Future directions include bridging the sim-to-real gap by either deploying these policies on state-of-the-art tendon-driven hardware or exploring methodologies to map virtual muscle activations (acting as variable stiffness/compliance signals) onto traditional electric-motor-driven humanoids equipped with whole-body impedance control.

O. Extended Context and Interpretation

This section retains background and interpretation that complement the concise related-work discussion in the main paper.

How the main findings use the evidence. T1 concerns a reversal between task-family averages: DynSyn-SAC leads DepRL by 40.00 percentage points in locomotion SR but trails it by 20.20 points in interaction. These values are differences of the displayed family means, not new experiments, uncertainty estimates, or zero-shot transfer scores. The reward and compression studies diagnose separate interventions and do not establish the cause of this reversal. T2 concerns the MuscleMimic–DepRL robustness comparison on five tasks; growing stair tracking error documents a failure but cannot distinguish reference-contact mismatch from all other possible causes.

Why success and EMG must be read together. T3 uses the existing common-metric table: stair adaptation changes SR from 0% to 100% and mean EMG correlation from 0.51 to 0.52, while running changes SR from 90% to 100% and correlation from 0.43 to 0.77. The stair result is not an equivalence test between correlations. In particular, the unsuccessful MM stair policy’s nonzero correlation shows that normalized waveform similarity does not certify task completion; it should not be read as equally successful or biologically equivalent behavior. Phase optimization and amplitude normalization further restrict the conclusion to waveform shape. A future causal account would need to separate changes in behavior, evaluation coverage, and individual adaptation components; the present paper does not claim that separation.

Musculoskeletal Models and Control. Musculoskeletal (MSK) agents differ from torque-driven humanoids: muscle excitations are transformed by activation-contraction dynamics into muscle forces and then mapped to joint torques through tendon paths and moment arms, introducing redundancy, delay, nonlinear muscle-tendon dynamics, and biomechanical constraints [25–27]. Classical platforms established workflows for anatomically grounded model construction, scaling, and motion-data analysis [25–28], while MyoSim and MyoSuite brought physiologically realistic MSK models into MuJoCo-style contact-rich RL

settings [1, 29–31]. Recent work further advances MSK control through model-based muscle controllers, hierarchical planning, markerless-capture imitation, and physical musculoskeletal humanoids [32–36], while full-body systems have scaled toward large-scale motion imitation, richer anatomy, and whole-body behavior emulation [12, 13, 20]. These advances motivate a systematic evaluation of full-body MSK control.

What the comparisons can establish. The full-suite comparison holds the primary body model fixed and changes task demands, allowing differences in success coverage and robustness to be assessed across stabilization, locomotion, and interaction. The focused studies then examine distinct limitations: reward reweighting changes optimization priorities, action compression trades control expressivity for lower-dimensional exploration, and imitation supplies reference structure that can become mismatched to contact. These studies do not use identical information: expert-trained representations and pretrained motion priors provide additional knowledge beyond task rewards. Accordingly, the reward-based leaderboard and the focused diagnostic studies should not be interpreted as one complete ranking over all five paradigms.

Residual adaptation and physiological interpretation. Residual learning adds corrective actions to an existing controller or skill policy [37–39]. In this paper it serves as a case study motivated by the benchmark’s contact and recruitment diagnostics. The walking, running, and stair results evaluate the combined residual, reward, and timing pipeline. They show task recovery and changes in measured recruitment, but do not isolate a causal benefit of each component. Similarly, phase-optimized EMG correlation characterizes normalized waveform agreement rather than absolute activation, absolute timing, or a unique human solution. Preserving these distinctions is necessary when relating task-level gains to physiological diagnostics.

P. DIRECT GPT-6 CONTROL: EXPERIMENTAL DETAILS

The main paper reports the aggregate outcomes of the direct-action diagnostic. This section records the controller boundary, state serialization, rollout procedure, stopping rule, and metric computation used to obtain those values; it does not add another performance table.

Controller-visible state.

At decision index k , GPT-6 Astra could read only two JSON files: static environment metadata (`env_info.json`) and the current serialized policy observation (`decision_k_controller_visible.json`). The metadata records the task, actuator names and ordering, action bounds, observation layout, simulator timesteps, motion-prior metadata, and decision budget. The observation is a lossless restructuring of the 2,418-dimensional float32 policy vector; flattening the serialized fields exactly reconstructs the original vector.

Observation block	Dim.
Root position without global x-y	5
Joint positions	82
Root velocity	6
Joint velocities	82
Muscle length, velocity, force, excitation, and activation for 354 muscles	1,770
Right/left foot and toe touch sensors	4
Goal and reference features	469
Total	2,418

TABLE XVIII. Controller-visible observation at each GPT-6 decision; dimensions are identical for Walk, Run, and Stairs.

The 469-dimensional goal block contains current relative positions, angles, and velocities for 16 tracked sites; current reference-site positions; four look-ahead blocks containing root-position delta, root-velocity delta, and site positions; and motion phase. The model could not inspect the saved trajectory, logging-only diagnostics, rendered frames, reference-controller outputs, future observations, or reference states beyond these policy-visible look-ahead features.

Action contract.

Each response was a bare JSON array whose index order follows the MuJoCo actuator ordering in the metadata. The runner required exactly 354 numeric, finite values, rejected Boolean entries, and checked every component against the environment bound $[-1, 1]$. Accepted arrays were cast to float32 and applied without clipping, smoothing, interpolation, optimizer refinement, baseline-action addition, or other post-processing. The base-policy callback was disabled, so the accepted array was the complete muscle-excitation command.

Actions were selected in the active GPT-6 Astra Codex session rather than through an external language-model API. The auditable controller record comprises the visible input files, accepted action arrays, and observation-to-action timestamps; hidden reasoning was not stored.

Pause, replay, and stepping protocol.

The interaction used a file-mediated paused session. A session reset the environment with seed 0 and exported decision 0. After one action file was submitted, the runner validated it, rebuilt the same seeded environment, deterministically replayed all committed actions, and applied the new action until its simulation-time deadline. If the rollout remained valid, it exported exactly one new observation and exited. This made each decision recoverable from the manifest and committed action sequence.

The physics timestep was 0.001 s and ten physics substeps formed one 0.01-s native control step. Action k was held with zero-order hold until simulation time $0.2(k+1)$, corresponding to 20 native control steps. Termination, truncation, and the fall gate were evaluated after every native step rather than only at GPT decision boundaries. Wall-clock selection delay did not advance simulated time and was not interpreted as model inference latency.

Fall gate and analyzed prefix.

Stopping and offline-analysis quantities were not controller-visible. For normalized root quaternion $q=(w,x,y,z)$, root tilt was $\arccos(\text{clip}(1 - 2(x^2 + y^2), -1, 1))$. A sample was marked fallen when generalized position or velocity was non-finite, root height was below 0.6 m, or root tilt exceeded 70 degrees. The current hold then stopped immediately. Metrics and videos use the inclusive prefix ending at the first fallen sample.

The immutable Walk audit log was produced before the stop gate was repaired and contains states after its first fallen sample at 0.75 s. Those post-fall states are excluded from every reported value and visualization.

Metric computation.

At each native step, site-tracking error is the square root of the mean squared coordinate difference between the simulated and reference positions over 16 tracked sites (48 coordinates). The reported tracking RMSE is the root mean square of this per-step series over the first-fall-truncated prefix. All native-control samples are included, not only GPT decision points.

First divergence uses a data-derived reference rather than a manually selected task threshold. A reference trajectory is generated with the official mm-fullbody-base MuscleMimic checkpoint and zero residual correction. Divergence is the first sample of the first five consecutive native-control samples for which GPT tracking error exceeds both the time-aligned reference error and the 95th percentile of all reference errors. Five samples correspond to 0.05 s. If the sustained condition is absent before termination, divergence is N/A. The reference rollout defines this timestamp only; it is not an extra result row in the direct control table.

Protocol scope.

The diagnostic uses one seed and a maximum 3-s simulated horizon for each task. It isolates the tested direct-action interface; it is not a statistical benchmark of language models and does not support a universal claim about GPT-6 control. In this configuration, the reasoner acts at 5 Hz while the musculoskeletal system is controlled at 100 Hz, so errors arising within a 0.2-s hold cannot be corrected before the next decision.

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